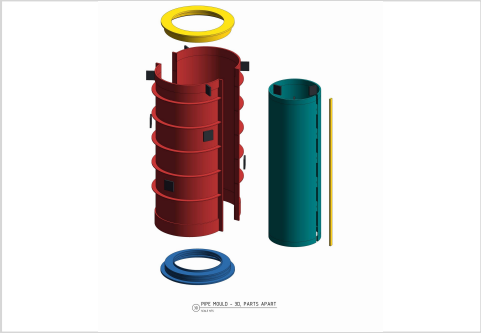
Civil & structural engineer, Riyadh

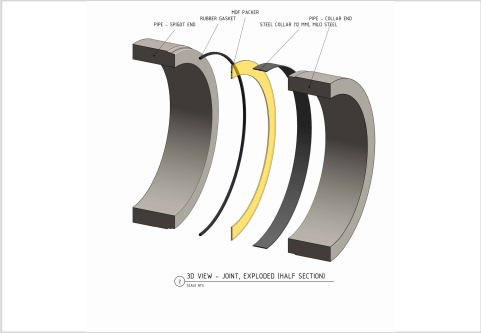
Riyadh, Saudi Arabia
+966 53 726 5921
eyashamadd@gmail.com
Jordanian national

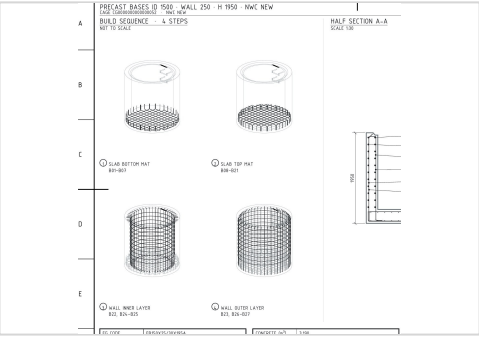
Eyas Hamad

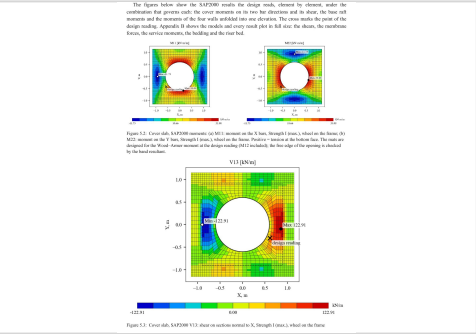
I design precast structures, and I build the systems that design, draw, price and make them.

Structural design to AASHTO, ASTM, ACI, EN and MOMRA; BIM and drawing automation; and the SAP manufacturing data behind each product, built at a Saudi precast manufacturer supplying the Kingdom's giga- and mega-projects.









Inquiry & pricing
~4,000
quotations since 2023, across Saudi giga- and mega-projects

Design
5
product families with design engines to ASTM, AASHTO and ACI

Drawings & BIM
6,394
Revit sheets for 1,345 manhole products

Costing
TABS
activity-based costing used with MAS

SAP product
~40,000
materials, 34,000+ BOMs and ~11,000 routings

Production
88
work centers tying routings to costing and planning

What I deliver

Tender and submittal work across NEOM, Qiddiya, Diriyah, King Salman Park, EXPO 2030, King Salman International Airport and KAFD.

Structural design of precast products

RC pipes, jacking pipes, circular and rectangular manholes and box culverts designed to AASHTO LRFD, ASTM, EN 1916, DWA-A 161, ACI and MOMRA, with SAP2000 finite-element models and complete calculation memos.

Precast technical office leadership

Tender analysis and pricing, product standards, submittals for consultant review, drawings and BBS, run as one controlled process from inquiry to production.

BIM and drawing automation

Revit product systems that generate native-rebar models, sheets, PDF/DWG/IFC, BBS and BOM from product data, so a full drawing library is a run, not a project.

Engineering ERP and product digitalisation

An inquiry-centred platform that takes each order from the client's BOQ to a production-ready SAP product, with the company's engineering rules and know-how in tested code.

SAP S/4HANA manufacturing master data

Material master, BOMs, routings, work centers and classification designed from an empty system to Go-Live, with engineering, costing, procurement, production and planning on one product model.

AI-augmented engineering delivery

Multi-agent AI coding workflows with cross-model review and automated test gates, used to build and maintain production engineering tools quickly and safely.

~4,000
quotations priced since 2023

5
precast product families automated

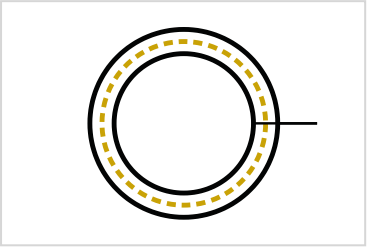
6,394
drawing sheets generated in Revit

~40,000
SAP materials designed to Go-Live

17,000+
automated tests behind the tools

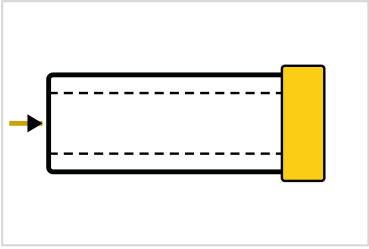
Five product families, automated end to end

Design, drawings and BIM are automated for five precast product families.



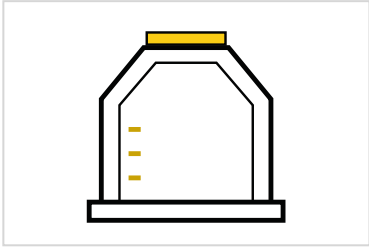
RC pipes

78-product standard range designed to ASTM C76 and AASHTO direct design; 67 standard drawing sets.



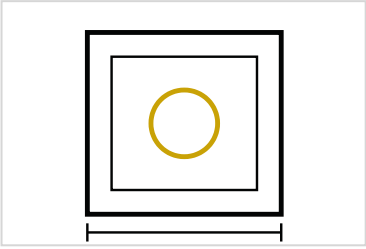
Jacking pipes

Design engine, calculation memos, catalogue and Revit drawing sets.



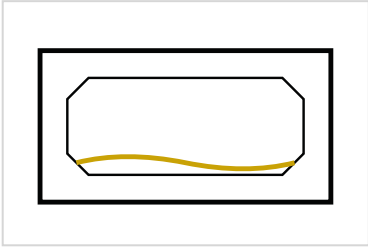
Circular manholes

Revit manhole product system v1.0, with a 6,394-sheet standard library for 1,345 manhole products.



Rectangular manholes

SAP2000 design engine; analysis cycle cut from 28+ minutes to about 6.



Box culverts

Single and twin cell designer v2.0: memos, BBS and drawings under 7,000+ tests.

Each family has its own design engine with the governing codes programmed in, so every configuration is checked against code before a memo, drawing or bill of materials is released.

Structural design capability

What each design engine covers, from the design basis of every product family.

Family	Codes	Loads	Analysis	Checks
RC pipes	ASTM C76M, C497; AASHTO LRFD 2024 §12.10; ACPA; MOMRA	Earth (vertical arching), fluid, HL-93 and MOMRA vehicles, dry and submerged	Indirect design (installation Types 1-4); direct Heger design on a closed ring; Eriksson Pipe and SAP2000 cross-checks	Flexure, crack control, diagonal tension, radial tension, steel limits, stirrups, flotation
Jacking pipes	BS EN 1916, BS 5911-1, DWA-A 161, ATV-DVWK-A 127, EN 1992-1-1	Earth by soil group, groundwater, traffic, jacking force with joint deflection	DWA-A 161 ring; packer stiffness; SAP2000 ring frame and 3D solid end-zone model	Ring flexure and crack width, fatigue, end-zone splitting and padding ties, collar and packer
Manholes	AASHTO LRFD 10th, ASTM C478M, ACI 318M-25, ACI 350, MOMRA live load	Earth (ACPA arching), at-rest pressure, HL-93 with surcharge, groundwater and buoyancy, handling	SAP2000 shell model per piece on compression-only joints and springs; Wood-Armer slab design	Biaxial flexure, hoop forces, shear, strut-and-tie and punching at openings, cracking, flotation, bearing
Box culverts	AASHTO LRFD 10th or MOMRA, ASTM C1433M / C1577, ACI 318M-25	Earth (Marston and trench factors), lateral earth, surcharge, water, HL-93, HS20, MOMRA vehicles	SAP2000 frame on compression-only Winkler springs; every vehicle placement evaluated	Flexure with thrust, shear, crack control, minimum steel, wall stability, bearing, flotation, lifting
Steel moulds	BS EN 10025 (S275JR), EN 1993-1-8, BS EN ISO 3834-3, ISO 5817, BS EN 1916 tolerances	Fresh-concrete pressure as a full liquid head with a factor of 1.5 for vibration; lifting	Part-by-part checks inside generated Revit mould sets	Lifting lugs, governing ratio per part, product and mould tolerances, shop checks before the first cast

RC pipes: a standard range designed to code

ASTM C76M Classes III to V, AASHTO LRFD indirect and direct design

A pipe range must carry earth, traffic and water loads in four installation types, prove its class in the three-edge-bearing test, and stay consistent across every size, class and cage.

What I built

- Design engine for ASTM C76M Classes III/IV/V to AASHTO LRFD 2024 §12.10: indirect design with installation Types 1-4, and direct (Heger) design on a closed-ring frame.
- Loads: earth, fluid, HL-93 and MOMRA vehicles with live-load distribution, dry and submerged states. Checks: flexure, crack control, diagonal tension, radial tension, steel limits, stirrups, flotation.
- Verified against Eriksson Pipe and a SAP2000 ring model; three-edge-bearing proof testing to ASTM C497 written into the inspection and test plan.
- Outputs per product: indirect and direct memos, a Revit drawing set, BBS and an 86-page catalogue.

Results

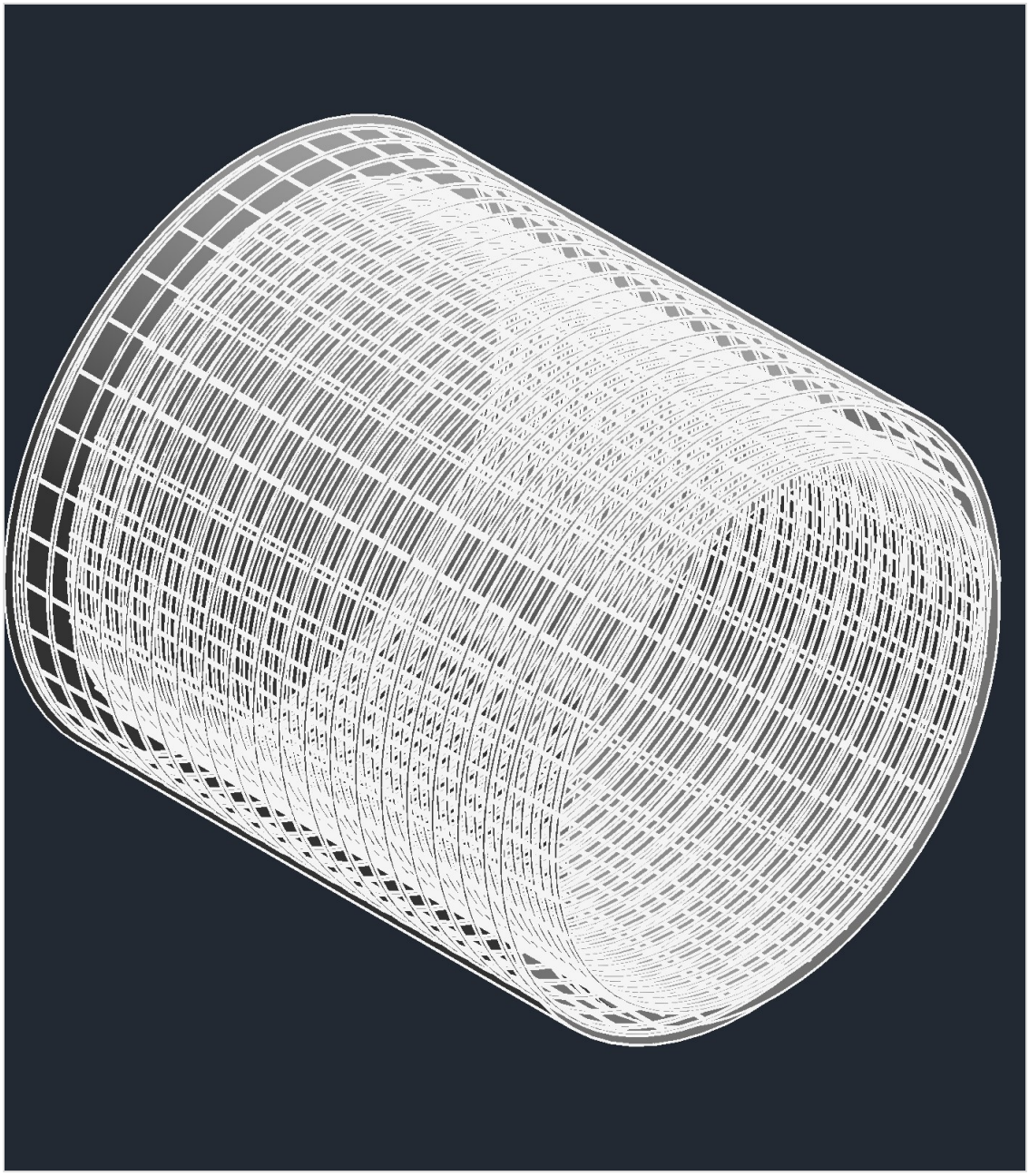
78
standard products, 26 moulds × 3 classes

0.04%
difference in the Eriksson Pipe cross-check

2,391
engine tests at release

86
page technical catalogue

Python design engine, SAP2000, Eriksson Pipe, Revit (C#), Excel



Revit model check of a Class IV RC pipe cage, generated from the design engine's output.

RC pipes: a standard range designed to code

CALCULATION MEMORANDUM — DIRECT DESIGN

Design summary

Table DS.1 Product and design basis

Item	Value
Product	Reinforced concrete pipe ID 1800 mm, wall B, Class IV (ASTM C76M-22a)
Wall h / OD / laying length	175 / 2,150 / 2,500 mm
Reinforcement	inner 1×T12@67 + outer 1×T12@113; no stirrups
Design method	AASHTO LRFD 2024 Art. 12.10.4.2 (direct design)
Installations	standard embankment Types 1–4 (Table 12.10.2.1-1)
Live load	MOMRA 600 kN (primary); HL-93 (secondary)
Fills / water	from 0.3048 m (1.0 ft); dry and submerged

Table DS.2 Governing case per check (highest D/C)

Check	Section	Case: Type · fill · live load · pipe · water	D/C	Result
Concrete shear	invert shear section	T1 · H 9.03 m · MOMRA 600 kN neglected at this fill · full · dry	1.000	PASS
Flotation	whole pipe	T1 · H 0.67 m · pipe empty · submerged, no live load	0.999	PASS
Maximum reinforcement, radial tension (4c-1)	invert	T1 · H 0.94 m · MOMRA 600 kN, 1 lane along · full · dry	0.950	PASS
Flexure	springline	T1 · H 0.94 m · MOMRA 600 kN, 1 lane along · full · dry	0.833	PASS
Crack control	invert	T4 · H 4.05 m · MOMRA 600 kN neglected at this fill · full · dry	0.762	PASS
Maximum reinforcement, compression (4c-2)	invert	T1 · H 0.94 m · MOMRA 600 kN, 1 lane along · full · dry	0.414	PASS
Minimum reinforcement	springline	T1 · H 0.94 m · MOMRA 600 kN, 1 lane along · full · dry	0.148	PASS

Verdict. Class IV passes the direct design check of AASHTO LRFD 2024 Art. 12.10.4.2 as designed within the fill ranges of Section 8 (dry and submerged); the highest D/C over the ranges is 1.000 (Concrete shear, invert shear section). The check adds no reinforcement. Production is released on the conditions of Section 9.

Direct-design memo summary to AASHTO LRFD 2024 §12.10.4.2: governing case and D/C for every check.

Annexes (continued)

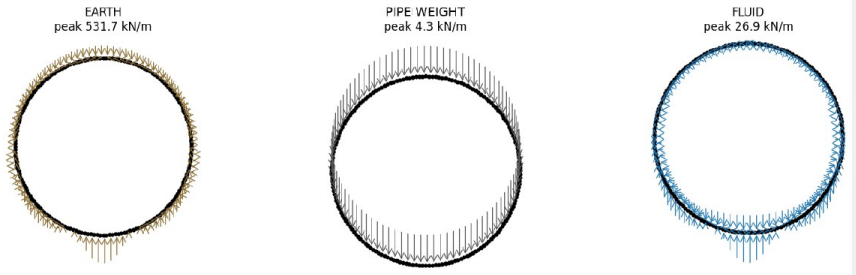


Figure B.1 Case 1: the loads on the ring as applied in the model (earth, pipe weight, fluid; unfactored, peak value per load)

Source: SAP2000 v26 model input

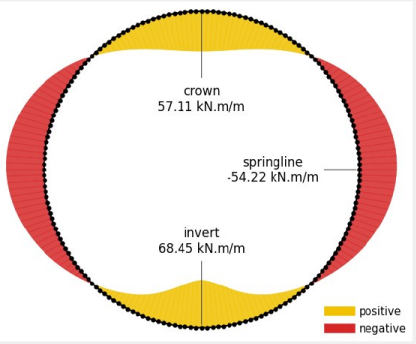


Figure B.2 Case 1: bending moment, factored combination (inside tension positive, drawn inward)

Source: SAP2000 v26 analysis output

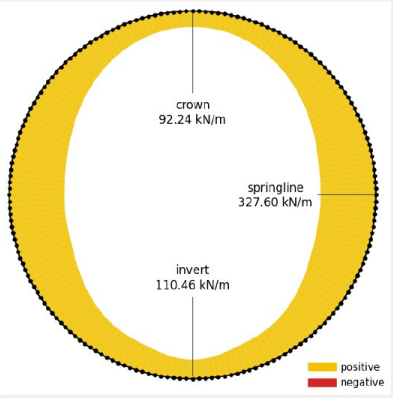


Figure B.3 Case 1: axial force, factored combination (compression positive, drawn inward)

Source: SAP2000 v26 analysis output

continued overleaf

Ring model loads, bending moment and axial force, cross-checked in SAP2000.

RC pipes: a standard range designed to code

FP1.8-2.15L2.5C4 - R.C PIPE Class IV ID 1800 MM L=2.5 M

BAR BENDING SCHEDULE

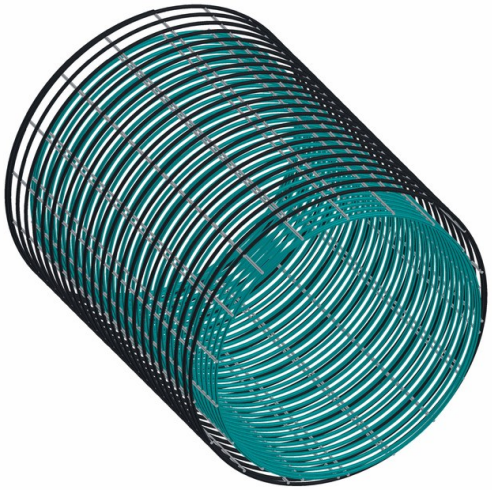
MARK	MEMBER	BAR	No.	LENGTH mm	BS CODE	SHAPE	A	B	MASS kg	SAP DESCRIPTION
OS1	OUTER CAGE	Ø12	1	156025	77	B A	2085	110	138.521	Steel Coil (12) mm CDW
IS1	INNER CAGE	Ø12	1	227675	77	B A	1860	65	202.133	Steel Coil (12) mm CDW
OL1	OUTER CAGE	Ø8	18	2450	00	A	2440		17.401	Cut Steel 8mm x 2.48m
IL1	INNER CAGE	Ø8	18	2450	00	A	2440		17.401	Cut Steel 8mm x 2.48m
	OUTER CAGE								155.922	
	INNER CAGE								219.534	
	TOTAL PER PIPE								375.456	

OS1 / IS1: SHAPE 77 HELIX, A OUTSIDE DIAMETER, B PITCH, C = 24.0 / 39.2 TURNS INCLUDING ONE FLAT CLOSING TURN AT EACH END. HELIX FROM 50 m CDW COILS; COIL CHANGES BUTT-WELDED, ASTM C76M 8.1.8.1 (PULL TEST >= 75 % OF MIN. TENSILE STRENGTH).
SP1 SPACER: CONCRETE BLOCK MADE IN PLANT, 25 x 25 WITH TIE WIRE, HEIGHT = CLEAR COVER (+2 / -0): 19 mm TO THE CORE (INNER CAGE), 33 mm TO THE JACKET (OUTER CAGE); ON A HOOP BETWEEN TWO LONGITUDINALS; 4 PER RING AT THE QUARTER POINTS; RINGS AT EACH CAGE END (<= 300 mm) AND AT MID-LENGTH. 24 No. PER PIPE.

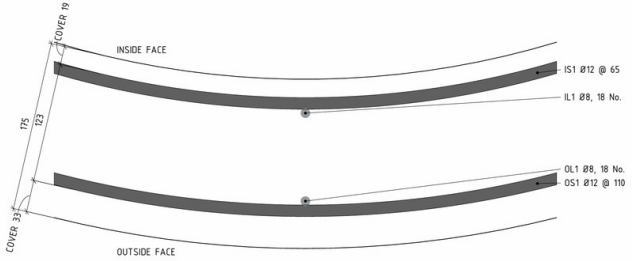
SAP MATERIALS (PER PIPE)

CODE	DESCRIPTION	PCS	kg PER PIPE
RSTCOIL-12	Steel Coil (12) mm CDW		340.654
RSTCUT08248	Cut Steel 8mm x 2.48m	36	35.228

CUT BARS: FULL 2.48 m PIECES AS ISSUED; COIL: NET BBS LENGTH.



3 3D VIEW - CAGES
SCALE NTS



7 DETAIL 7 - WALL OF SECTION B-B AT THE INVERT
SCALE 1:2

LEGEND

KEY	MARK	MEMBER	BBS
	OS1	OUTER CAGE	Ø12 1 No. L 156.025 m 138.521 kg
	IS1	INNER CAGE	Ø12 1 No. L 227.675 m 202.133 kg
	OL1	LONGITUDINALS	Ø8 18 No. L 2.450 m 17.401 kg
	IL1	LONGITUDINALS	Ø8 18 No. L 2.450 m 17.401 kg

Standard drawing set, reinforcement sheet: 3D cage, joint wall detail and bar bending schedule.

RC pipes: a standard range designed to code

4.19 ID 1,800 mm

Wall 175 mm (Wall B) · outside diameter 2,150 mm · laying length 2,500 mm · tongue and groove joint, gap 5 – 15 mm, home at 12 mm · 6.767 t and 2.707 m³ per pipe

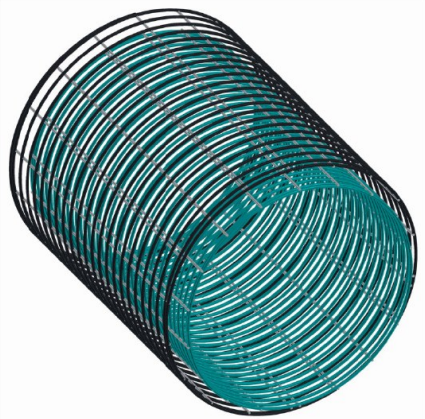


Figure 4.16 ID 1,800 Class IV: the continuous spiral cages

Source: JDCO standard drawing RCP-1800-IV-B-R01

Table 4.32 ID 1,800: Classes III, IV, V

	Class III	Class IV	Class V
SAP code	FP1.8-2.15L2.5C3	FP1.8-2.15L2.5C4	FP1.8-2.15L2.5C5
D-load 0.3 mm / ultimate, N/m/mm	65 / 100	100 / 150	140 / 175
Test load 0.3 mm / ultimate, kN/m	117.0 / 180.0	180.0 / 270.0	252.0 / 315.0
Concrete f'c, MPa	27.6	34.5	41.4
Inner cage, mm	Ø12 @ 108	Ø12 @ 67	Ø12 @ 46
Outer cage, mm	Ø10 @ 119	Ø12 @ 113	Ø12 @ 112
Steel area inner / outer, cm²/m	10.47 / 6.60	16.88 / 10.01	24.59 / 10.10
Stirrups	—	—	—
Steel per pipe, kg	253.9	371.7	459.1
ASTM C76M-22a basis	Table 3, Wall B, ID 1800	Table 4, Wall B, ID 1800	Special design, 7.2.3
Drawing set	RCP-1800-III-B	RCP-1800-IV-B	RCP-1800-V-B
Design memoranda	MEMO-RCP-1800-B-IND-III-R00 MEMO-RCP-1800-B-DIR-III-R00	MEMO-RCP-1800-B-IND-IV-R00 MEMO-RCP-1800-B-DIR-IV-R00	MEMO-RCP-1800-B-IND-V-R00 MEMO-RCP-1800-B-DIR-V-R00

Table 4.33 ID 1,800: fill height H in metres, dry installation

	Live load MOMRA 600 kN			Live load HL-93		
Installation	III	IV	V	III	IV	V
Type 1	1.1 – 6.9	0.7 – 10.9	0.3 – 15.4	0.3 – 6.7	0.3 – 10.8	0.3 – 15.4
Type 2	1.1 – 5.3	0.3 – 8.4	0.3 – 12.0	0.3 – 5.1	0.3 – 8.3	0.3 – 11.9
Type 3	1.3 – 4.1	0.3 – 6.5	0.3 – 9.3	0.3 – 3.9	0.3 – 6.4	0.3 – 9.2
Type 4	2.5 – 2.9	0.7 – 4.7	0.3 – 6.8	0.3 – 2.7	0.3 – 4.6	0.3 – 6.7

Submerged installations, the other diameters and the selection charts: Section 7.

Technical catalogue page generated from the engine: dimensions, cages and fill heights per class.

6.2 The four installation types

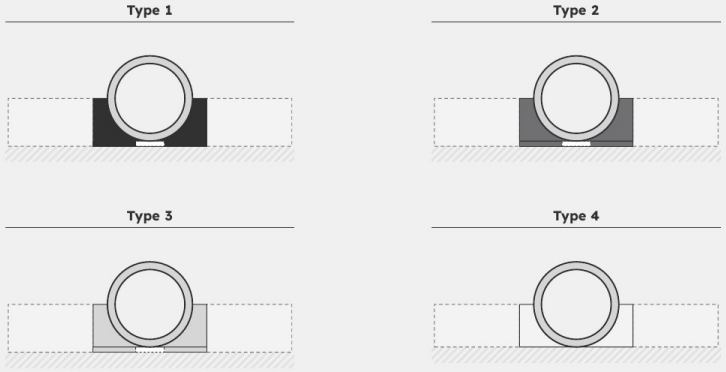


Figure 6.3 Installation Types 1 to 4: darker shading marks the higher compaction of the haunch, outer bedding and bedding

Source: zones of AASHTO LRFD 2024 Tables 12.10.2.1-1 and -2; zone widths after ASTM C1479M-07a Figs. 2 and 3; drawn to scale for ID 1,000

Table 6.1 The standard installations compared

					Maximum fill m, MOMRA		
Type	In short	Bedding thickness	Haunch and outer bedding	Lower side	Class III	Class IV	Class V
Type 1	highest quality: select granular soil, the most compaction and control	Do/24, ≥ 75 mm (3.0 in); rock: Do/12, ≥ 150 mm	95 % Category I	90 % Cat. I, 95 % Cat. II or 100 % Cat. III	6.1 – 7.6	9.9 – 11.8	14.2 – 16.6
Type 2	native silty granular soils allowed with compaction; no clay in the haunch	as Type 1	90 % Cat. I or 95 % Cat. II	85 % Cat. I, 90 % Cat. II or 95 % Cat. III	4.9 – 5.8	7.9 – 9.0	11.4 – 12.7
Type 3	lower compaction and less stringent inspection	as Type 1	85 % Cat. I, 90 % Cat. II or 95 % Cat. III	85 % Cat. I, 90 % Cat. II or 95 % Cat. III	3.7 – 4.5	6.1 – 7.1	8.8 – 10.0
Type 4	minimal soil and compaction requirements; no bedding unless on rock	soil: none; rock: Do/12, ≥ 150 mm	no compaction, except Cat. III at 85 %	no compaction, except Cat. III at 85 %	2.9 – 3.0	4.1 – 4.8	5.8 – 6.9

Requirements per AASHTO LRFD 2024 Table 12.10.2.1-1 (embankment; trench: Table 12.10.2.1-2, the lower side may be natural soil of equal firmness). Percentages are of the standard Proctor density (AASHTO T 99 / ASTM D698). Maximum fill: the range over all inside diameters of the highest fill of the class in the tables of Section 7 (MOMRA live load, dry).

continued overleaf

Standard installations Types 1-4 compared, with maximum fill per class.

RC pipes: a standard range designed to code

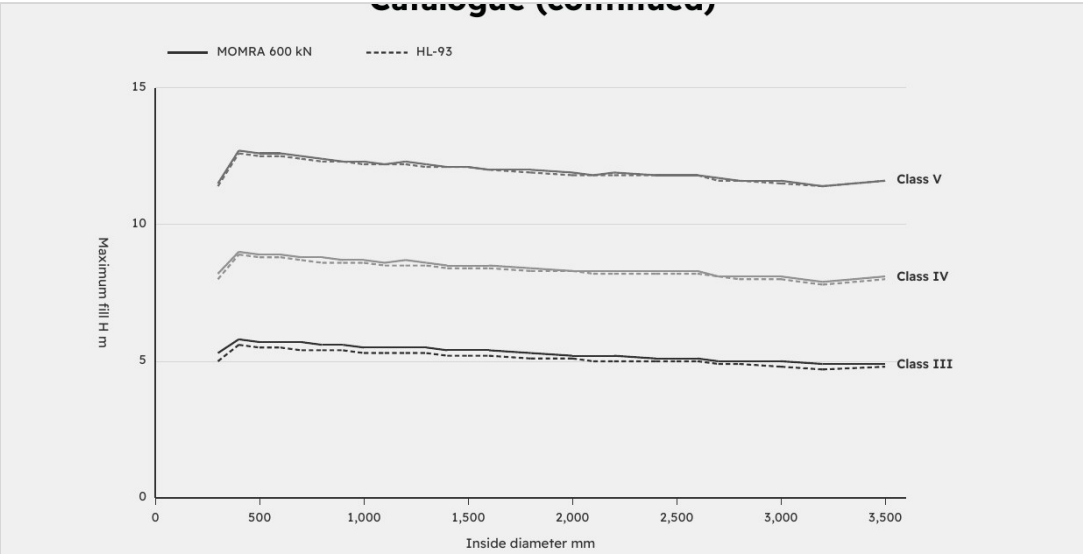
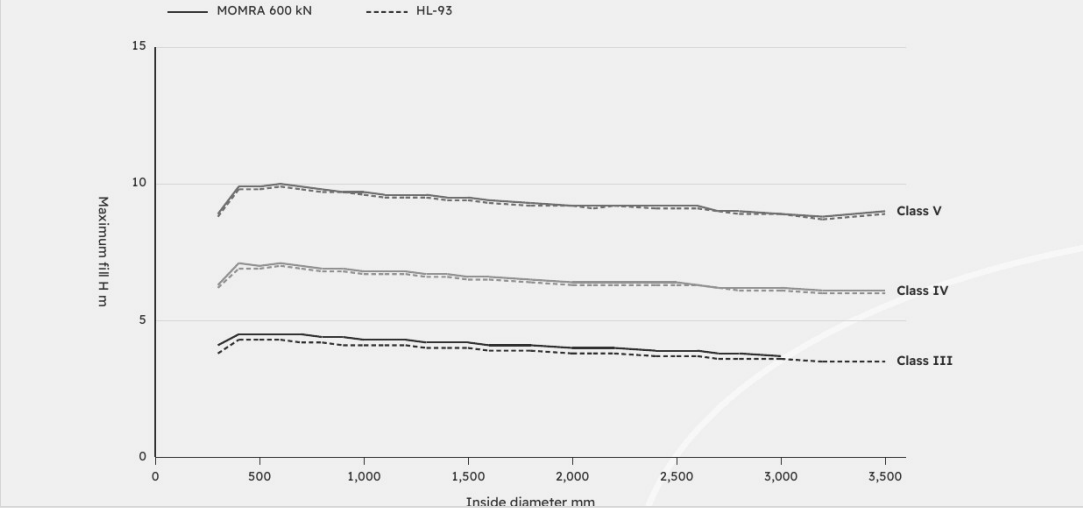


Figure 7.10 Maximum fill by inside diameter, installation Type 2

Source: the fill-height tables of this section



Maximum fill against inside diameter for each class and installation type.

Manholes: circular and rectangular, as shell models

AASHTO LRFD 10th, ASTM C478M and MOMRA live load in SAP2000

Buried manholes carry earth, traffic, groundwater and handling loads through stacked pieces whose joints take compression only; covers carry eccentric openings, and every depth type needs its own design.

What I built

- Each precast piece modelled as a SAP2000 shell with compression-only joints, bedding and base springs; loads cascade from the cover down to the base.
- Loads: earth with ACPA arching, at-rest pressure on effective stress, HL-93 with MOMRA surcharge, groundwater with buoyant soil weight, handling with a dynamic factor.
- Checks: biaxial flexure on every face, hoop compression and tension, shear, strut-and-tie and punching at cover openings, crack control, flotation at every joint plane, bearing, joint sliding and lifting.
- Slabs designed by Wood-Armer; every printed check traced to its SAP2000 reading in a 44-54-page memo.

Results

4 of 4
release cases with zero failed checks

3,437
tests on the release gate

28+ → ~6 min
rectangular analysis cycle

Python, SAP2000 API, Excel control workbook, Word/PDF memo generator

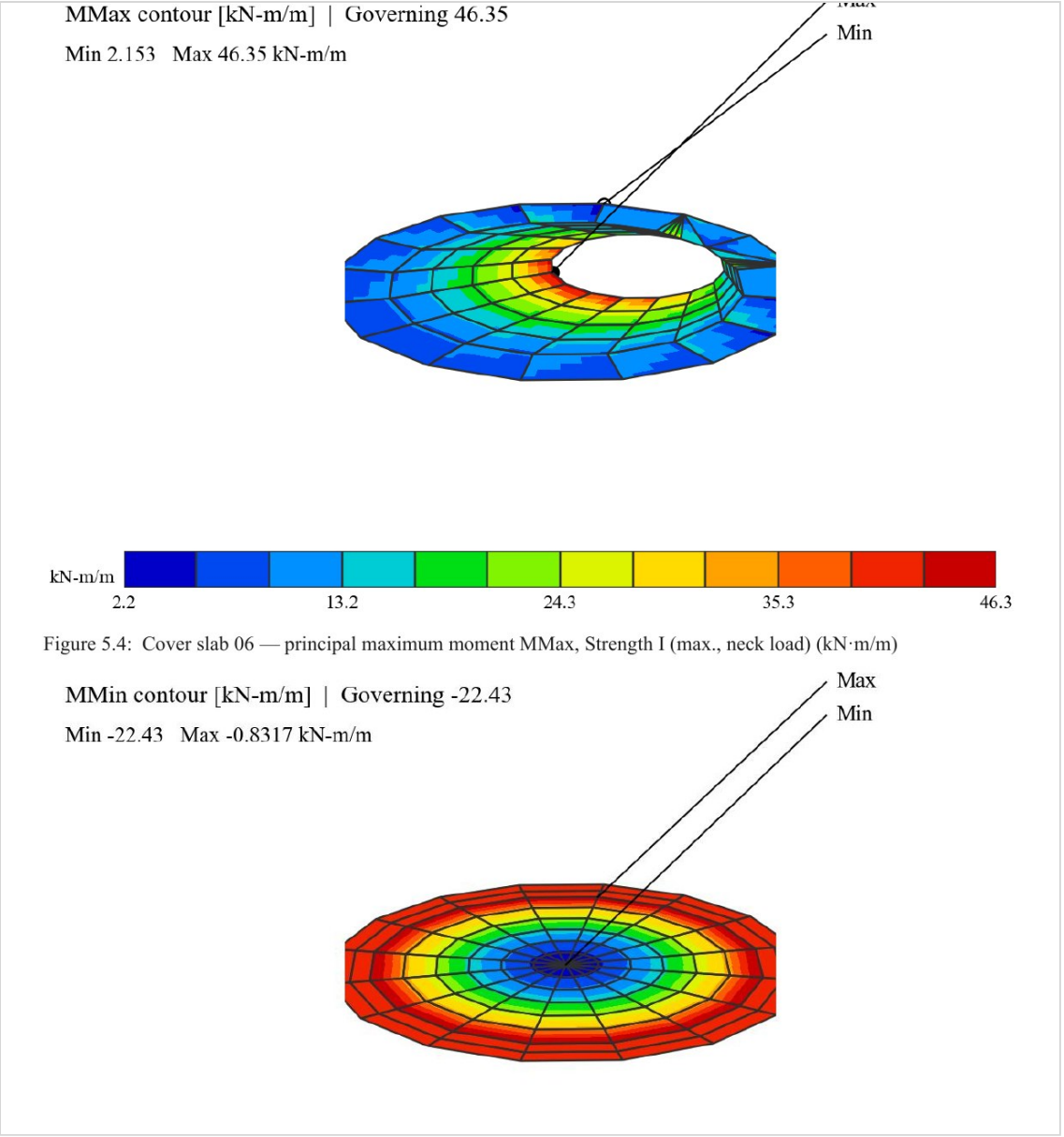


Figure 5.4: Cover slab 06 — principal maximum moment MMax, Strength I (max., neck load) (kN·m/m)

Circular manhole cover slab and base: principal moment contours under Strength I.

Manholes: circular and rectangular, as shell models

The figures below show the SAP2000 results the design reads, element by element, under the combination that governs each: the cover moments on its two bar directions and its shear, the base raft moments and the moments of the four walls unfolded into one elevation. The cross marks the point of the design reading. Appendix B shows the models and every result plot in full size: the shears, the membrane forces, the service moments, the bedding and the riser bed.

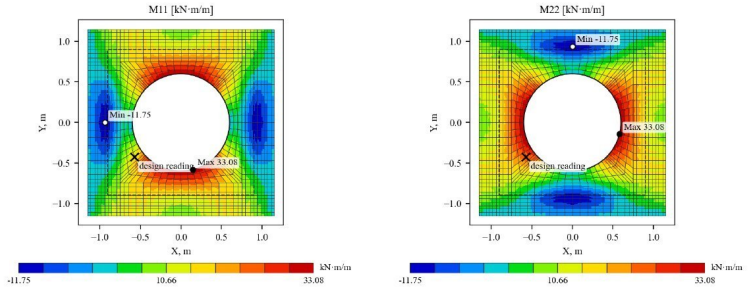


Figure 5.2: Cover slab, SAP2000 moments: (a) M11: moment on the X bars, Strength I (max.), wheel on the frame; (b) M22: moment on the Y bars, Strength I (max.), wheel on the frame. Positive = tension at the bottom face. The mats are designed for the Wood–Armer moment at the design reading (M12 included); the free edge of the opening is checked by the band resultant.

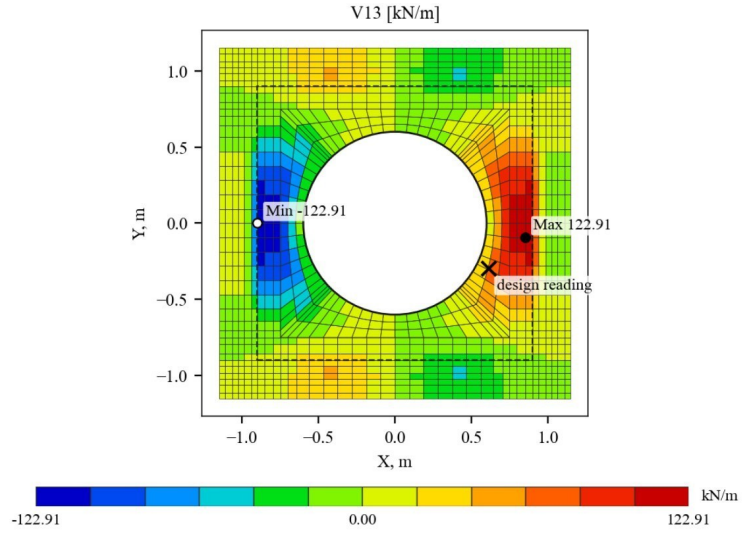


Figure 5.3: Cover slab, SAP2000 V13: shear on sections normal to X, Strength I (max.), wheel on the frame

Rectangular manhole cover with an eccentric opening: SAP2000 shell moments and shear, designed by Wood-Armer and strut-and-tie.

B.3. WALLS

The four walls of each piece are unfolded into one elevation (faces −Y, +X, +Y, −X from the left) and the pieces are shown at their heights in the stack; each piece is drawn under the combination of its own design reading.

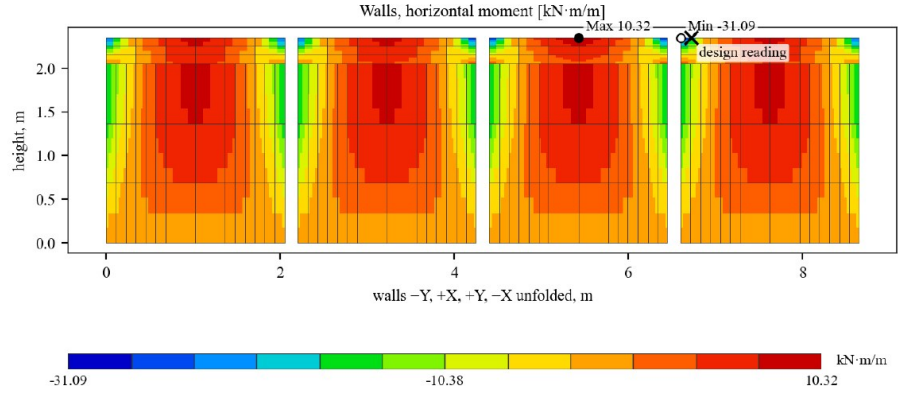


Figure B.7: Walls unfolded, Strength I, horizontal moment (positive = tension at the inner face); riser 1: Strength I (max.); base: Strength I (max.), compression-only bedding.

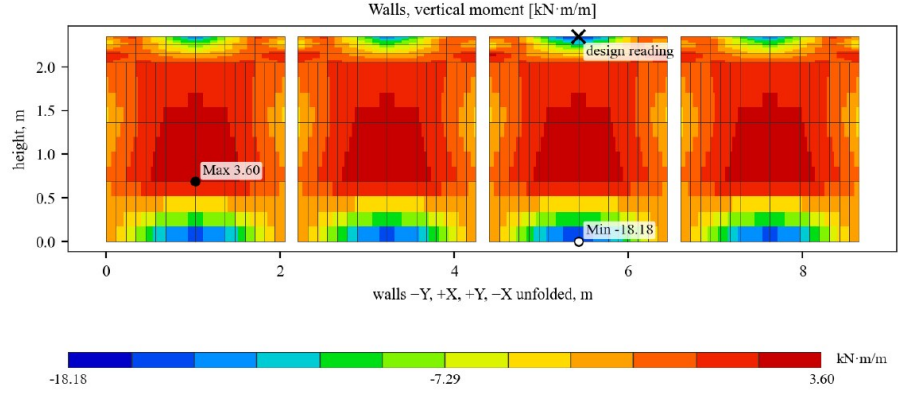


Figure B.8: Walls unfolded, Strength I, vertical moment (positive = tension at the inner face); riser 1: Strength I (max.);

Walls unfolded per piece: horizontal and vertical moments from the SAP2000 shell model.

Manholes: circular and rectangular, as shell models

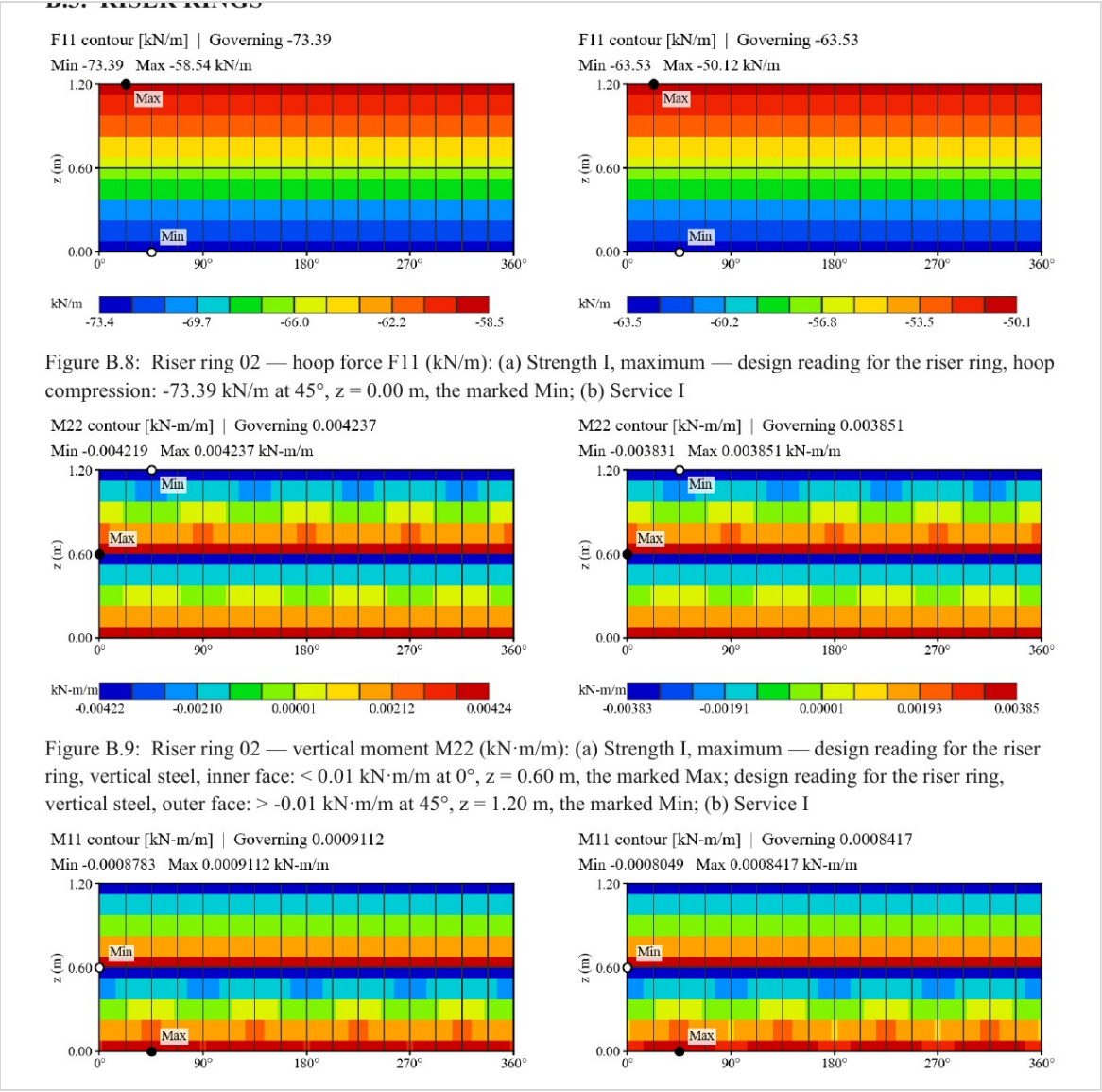


Figure B.8: Riser ring 02 — hoop force F11 (kN/m): (a) Strength I, maximum — design reading for the riser ring, hoop compression: -73.39 kN/m at 45°, z = 0.00 m, the marked Min; (b) Service I

Figure B.9: Riser ring 02 — vertical moment M22 (kN·m/m): (a) Strength I, maximum — design reading for the riser ring, vertical steel, inner face: < 0.01 kN·m/m at 0°, z = 0.60 m, the marked Max; design reading for the riser ring, vertical steel, outer face: > -0.01 kN·m/m at 45°, z = 1.20 m, the marked Min; (b) Service I

Riser ring: hoop force and moments, every face and direction checked.

BIM product system for manholes

From a design order to native-rebar models and issued sheets

Manhole drawings were drafted case by case, so the same product could appear in different forms and every new size meant new drafting.

What I built

- A Revit 2027 add-in driven by a C# rule engine: families, native rebar, openings, steps, views and sheets, exported to PDF, DWG, IFC and RVT with BBS, BOM and take-off.
- One master data set feeds every sheet; an automated check holds each sheet to the drafting standard.
- The same pipeline issues the RC pipe and jacking-pipe drawing sets.
- Released as v1.0 in October 2026 with 1,000+ automated .NET tests.

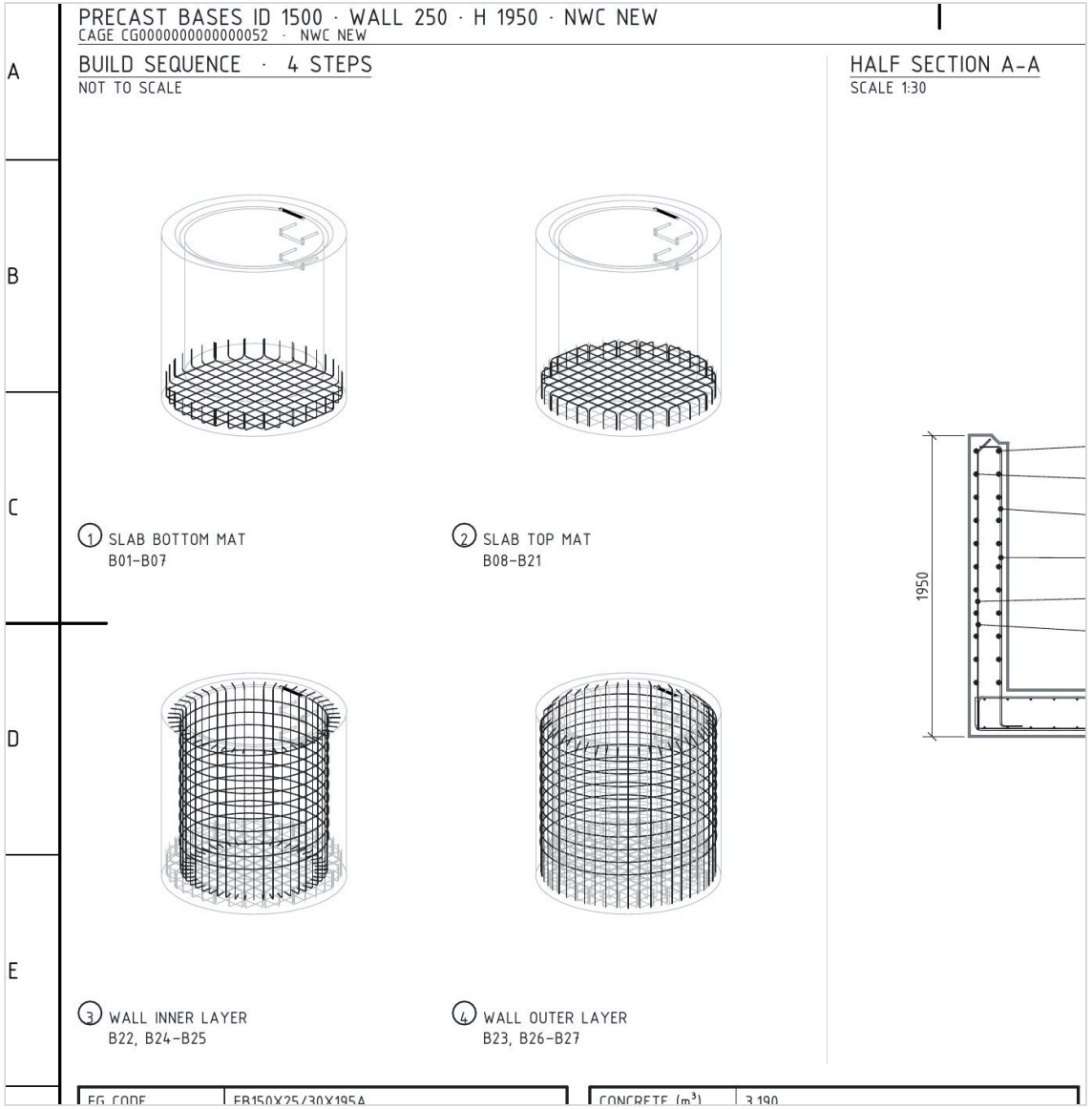
Results

6,394
sheets in the standard library

1,345
manhole products covered

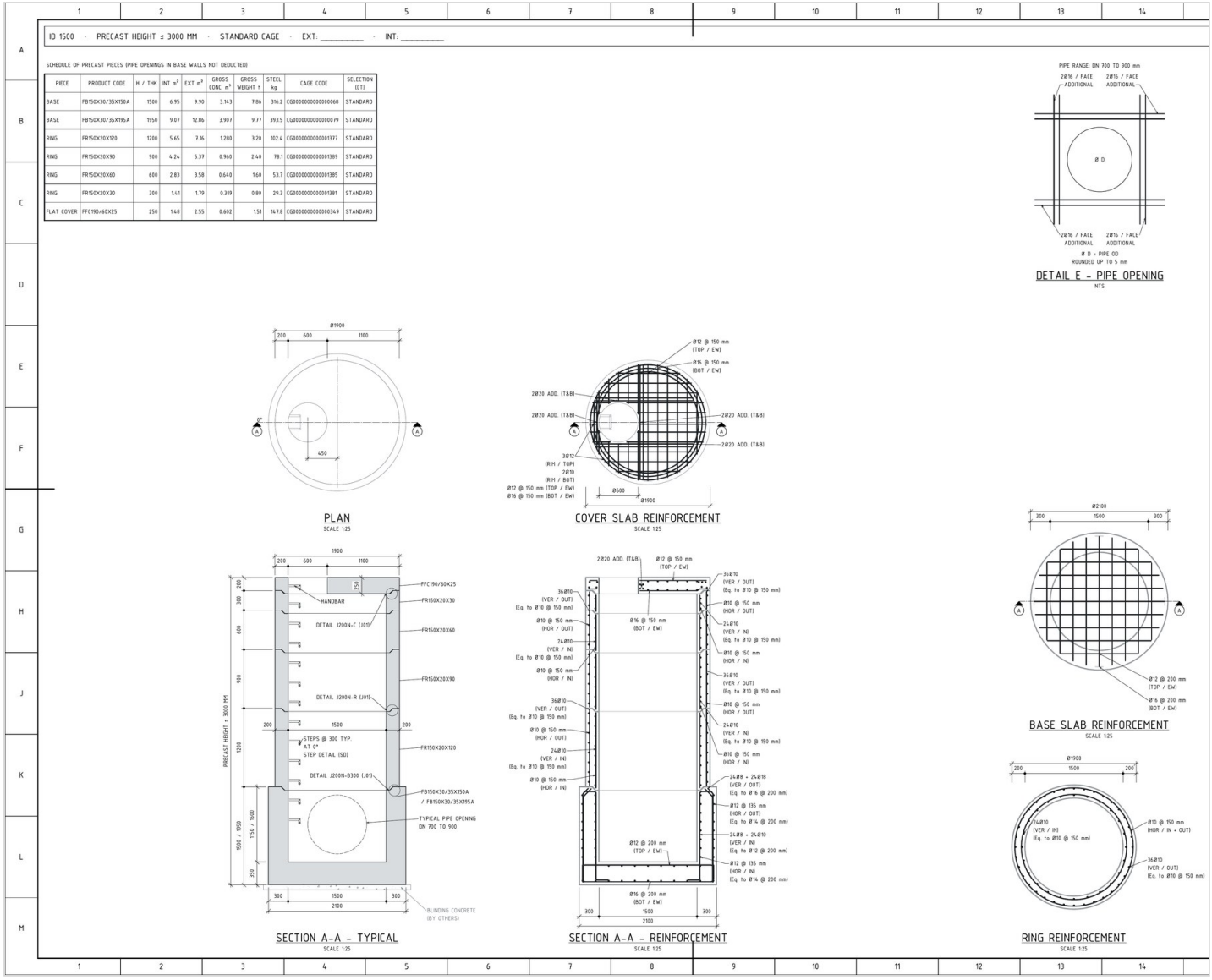
3,059
manhole types in the library

Revit 2027 API (C#), .NET, Python, Excel Designer



Cage-library sheet: native rebar in four build steps with half section, for the fabrication shop.

BIM product system for manholes



Revit general-arrangement sheet: stack section, cover, base and ring reinforcement, opening detail and piece table.

BIM product system for manholes

GENERAL DRAWING

BASE FB120X25/30X150A

BS 8666:2020 · CAGE CG0000000000000017 · STANDARD · 249.53 kg PER CAGE

MEMBER	BAR MARK	TYPE & SIZE	No. OF MBRS	No. IN EACH	TOTAL No.	LENGTH (mm)	SHAPE CODE	A	B	C	D	E/R	MASS (kg)
FB120X25/30X150A	B01	A615-420 Ø16	1	2	2	1980	21	210	1620	210			6.25
	B02		1	4	4	1925	21	210	1570	210			12.18
	B03		1	4	4	1765	21	210	1405	210			11.15
	B04		1	4	4	1440	21	210	1080	210			9.09
	B05		1	4	4	485	21	210	130	210			3.08
	B06	A615-420 Ø12	1	1	1	1975	21	215	1595	215			1.75
	B07		1	1	1	1950	21	200	1595	200			1.73
	B08		1	2	2	1925	21	215	1545	215			3.42
	B09		1	2	2	1900	21	200	1545	200			3.38
	B10		1	2	2	1755	21	215	1380	215			3.12
	B11		1	2	2	1735	21	200	1380	200			3.08
	B12		1	2	2	1425	21	215	1045	215			2.53
	B13		1	2	2	1400	21	200	1045	200			2.49
	B14	A1064 Ø12	1	1	1	41360	99	Ø1315	135	10			36.77
	B15		1	1	1	50530	99	Ø1610	135	10			44.92
	B16	A615-420 Ø8	1	24	24	1570	11	115	1475				14.90
	B17	A1064 Ø12	1	12	12	1575	11	125	1470				16.78
	B18	A615-420 Ø8	1	24	24	1535	11	1435	115				14.54
	B19	A615-420 Ø16	1	24	24	1540	11	1430	140				58.37

SAP MATERIAL	DESCRIPTION	Ø	BAR MARKS	No.	L (m)	kg
RSTCUT08350	Cut Steel 8mm x 3.50m	8	B16, B18	48	74.52	29.44
RSTCOIL-12	Steel Coil (12) mm CDW	12	B14-B15, B17	14	110.79	98.47
RSTDEF-12	Steel Bar (12mmx12m)	12	B06-B13	14	24.21	21.50
RSTDEF-16	Steel Bar (16mmx12m)	16	B01-B05, B19	42	63.38	100.12
TOTAL PER CAGE	CG0000000000000017			118	272.90	249.53

BS 8666:2020. VERIFIED SHAPES: A-E OUTSIDE mm. SHAPE 99 POLYLINES: A-E C.L. SEGMENTS, F ONWARD IN MARK NOTE. TURNS IN PATH ORDER. HOOPS / ARCS / SPIRALS: MARK NOTE.

B14-B15: A = FORMING C.L. Ø; B = PITCH mm; C = TURNS, THE FLAT END TURNS INCLUDED. ONE CONTINUOUS WIRE, ONE FLAT TURN AT EACH END; LAP = 0.

Bar bending schedule to BS 8666:2020 generated with the sheet, with SAP material lines.

Box culverts: a frame on Winkler soil

AASHTO LRFD 10th or MOMRA, ASTM C1433M / C1577, ACI 318M-25

A box culvert's design depends on fill, vehicle and soil stiffness together; single and twin cells, two code routes and the handling of heavy units all have to be covered.

What I built

- Plane frame with haunches on compression-only Winkler springs, run in SAP2000 through its API; a soil-contact solution is accepted only after equilibrium and complementarity checks.
- Loads: earth with Marston and trench factors, lateral earth, surcharge, water, and HL-93, HS20 or MOMRA vehicles with dynamic allowance; every generated vehicle placement evaluated.
- Checks: flexure with thrust, shear, crack control, minimum steel, wall stability, bearing, flotation and lifting at each handling stage.
- Package: memo, clause-by-clause compliance matrix, BBS, shop drawings and lifting design, verified by an independent frame, an independent contact checker and a truth set of published cases.

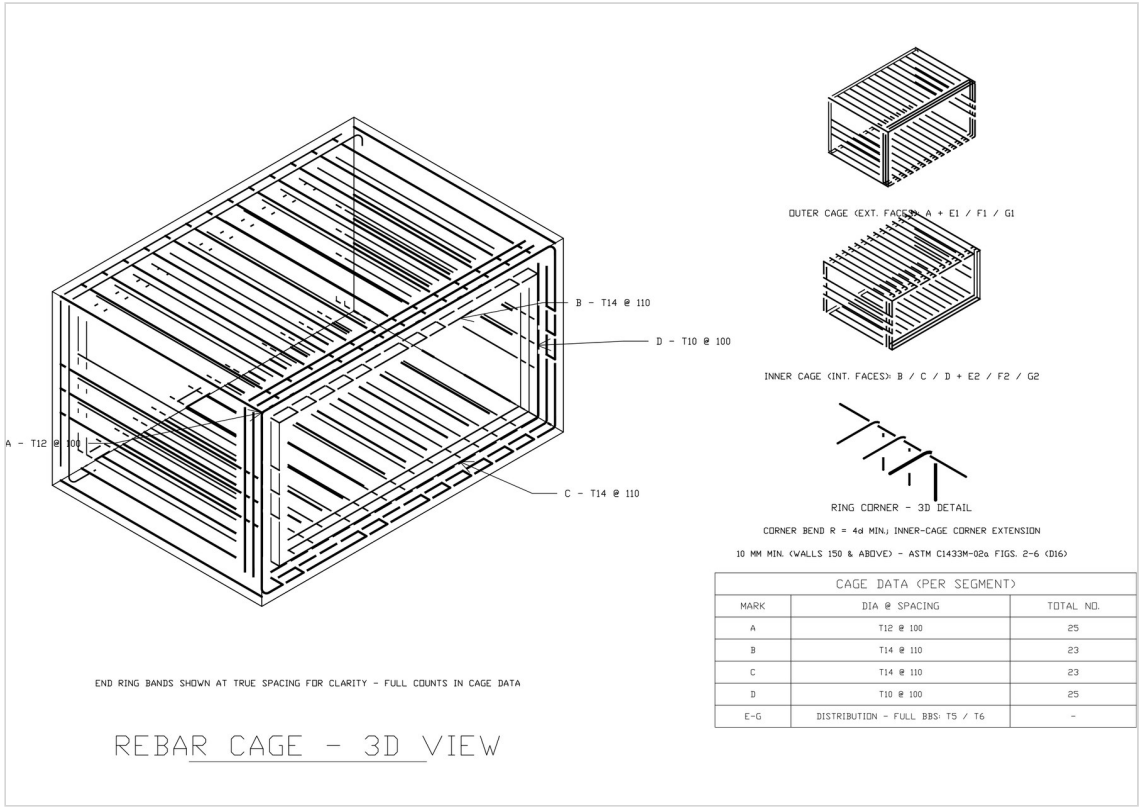
Results

v2.0
single and twin cell designer

7,000+
tests on the release gate

2
code routes: AASHTO and MOMRA

Python, SAP2000 API, Eriksson Culvert, Excel, local web UI



Shop drawing: 3D rebar cage with outer and inner cages, corner detail and cage data.

Box culverts: a frame on Winkler soil

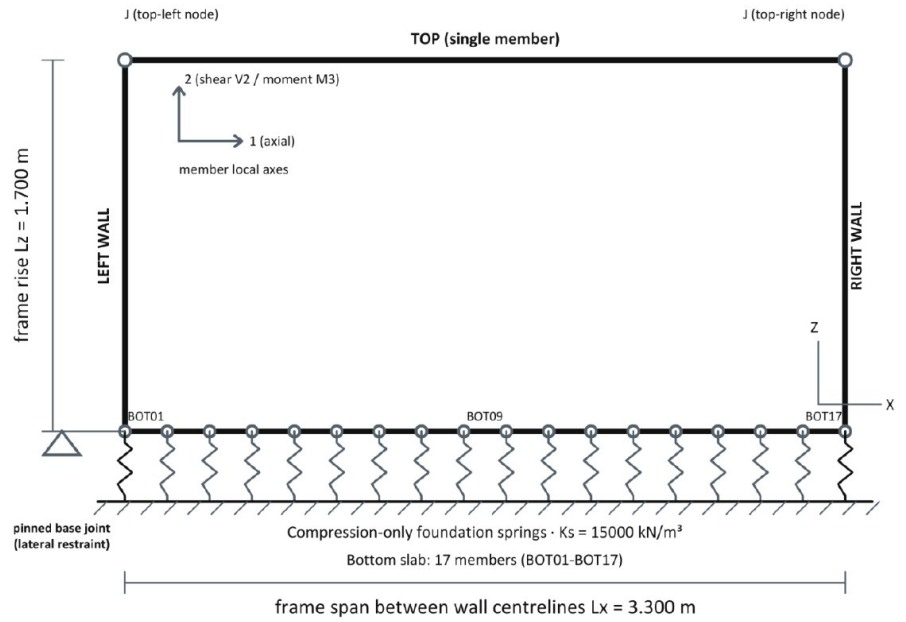


Figure 6.1 — Frame model & Winkler supports

6.2 MEMBER SECTIONS

Rectangular sections per 1 m strip: top slab 300 mm, bottom slab 300 mm on the Winkler subgrade springs, and both walls 300 mm. Self weight generated within the model.

WINKLER SUPPORT PARAMETERS

Table 6.1 — WINKLER SUPPORT PARAMETERS

Item	Value	Basis
Subgrade modulus Ks	$Ks = 15{,}000\ kN/m^3$	Project geotechnical input
Joint tributary width	$dx = 0.194\ m\ (17\ segments)$	Bottom-slab discretisation
Interior-joint spring	$k = 2912\ kN/m$	Ks x dx

Frame model on compression-only Winkler springs, with the support parameters it is solved with.

Table 6.3 — At H,min

Section	|Mu| (kN·m)	|Vu| (kN)	|Ms| (kN·m)
top slab — positive-moment region	98.023	157.464	63.872
bottom slab — left exterior support face	53.561	212.462	35.933
bottom slab — right exterior support face	53.561	212.462	35.933
bottom slab — positive-moment region	98.297	156.453	64.654
left wall — bottom support face	78.896	46.858	52.291
left wall — top support face	77.84	40.546	51.166
right wall — bottom support face	78.896	46.858	52.291
right wall — top support face	77.84	40.546	51.166
left wall — outer face, moment peak	78.896	46.858	52.291
right wall — outer face, moment peak	78.896	46.858	52.291

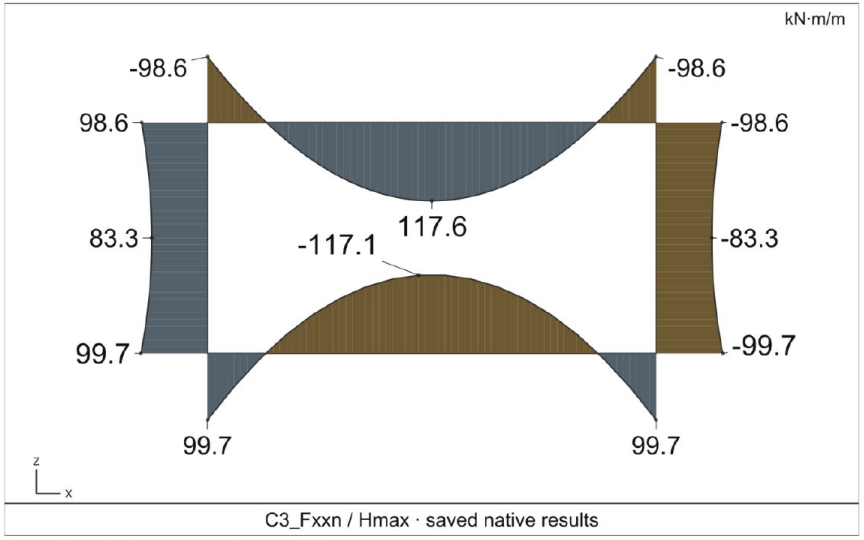


Figure 6.2 — Bending-moment diagram (M3)

Design actions per section and the bending moment diagram read back from SAP2000.

Box culverts: a frame on Winkler soil

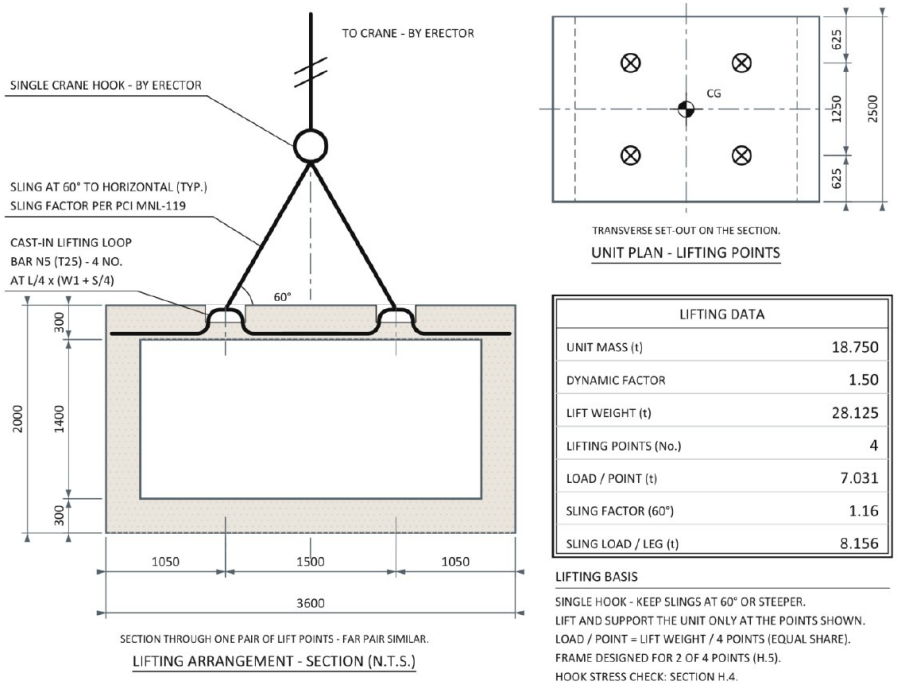
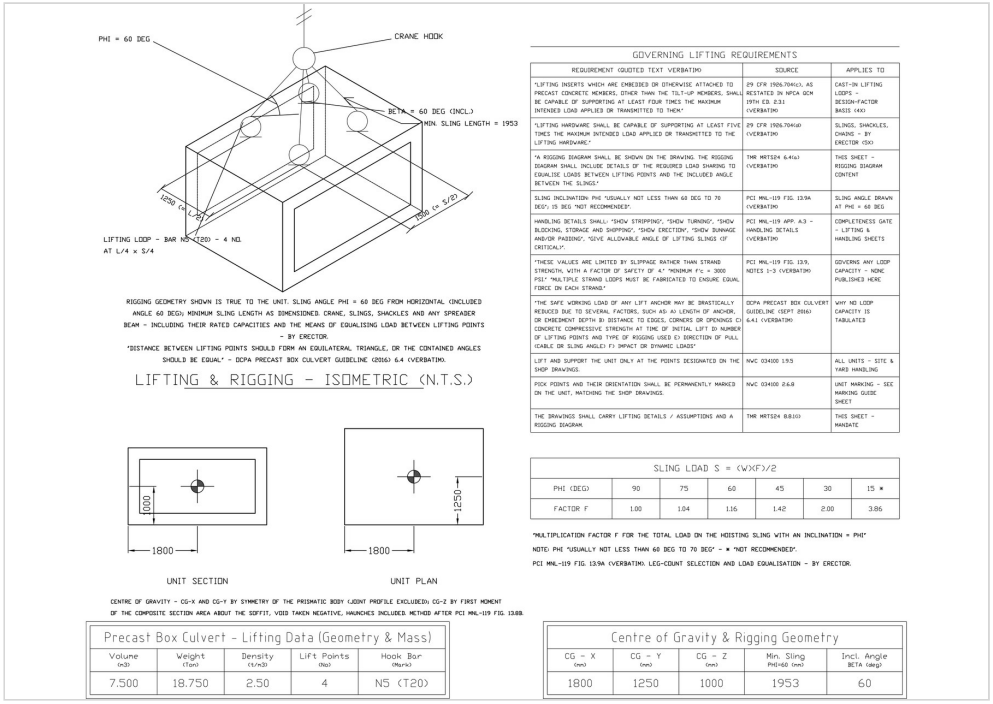


Figure H.1 — Rigging & sling geometry

H.5 LIFT-FRAME STRESSES

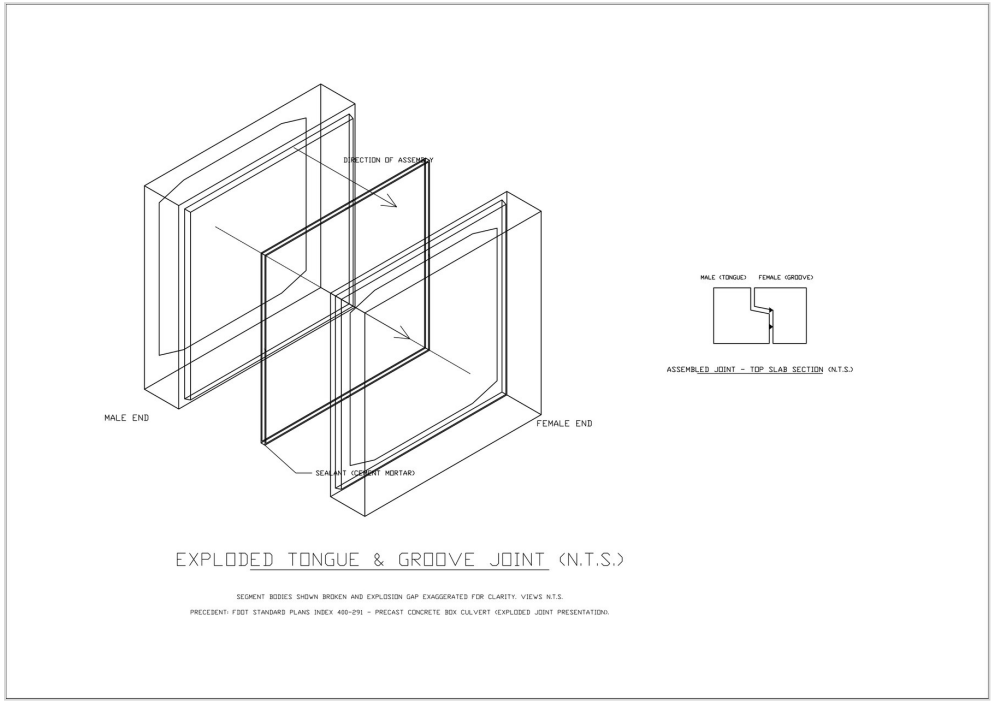
Two checks govern the lift: (a) the cast-in hooks are designed for the erection stresses of both conditions (§H.4), and (b) when the box hangs from the four points the walls and slabs bend — analysed in SAP2000 under self-weight × the dynamic factor and compared with the handling-stage capacity: the Section 7

Lifting design: rigging and sling geometry, lifting points and release-strength checks.

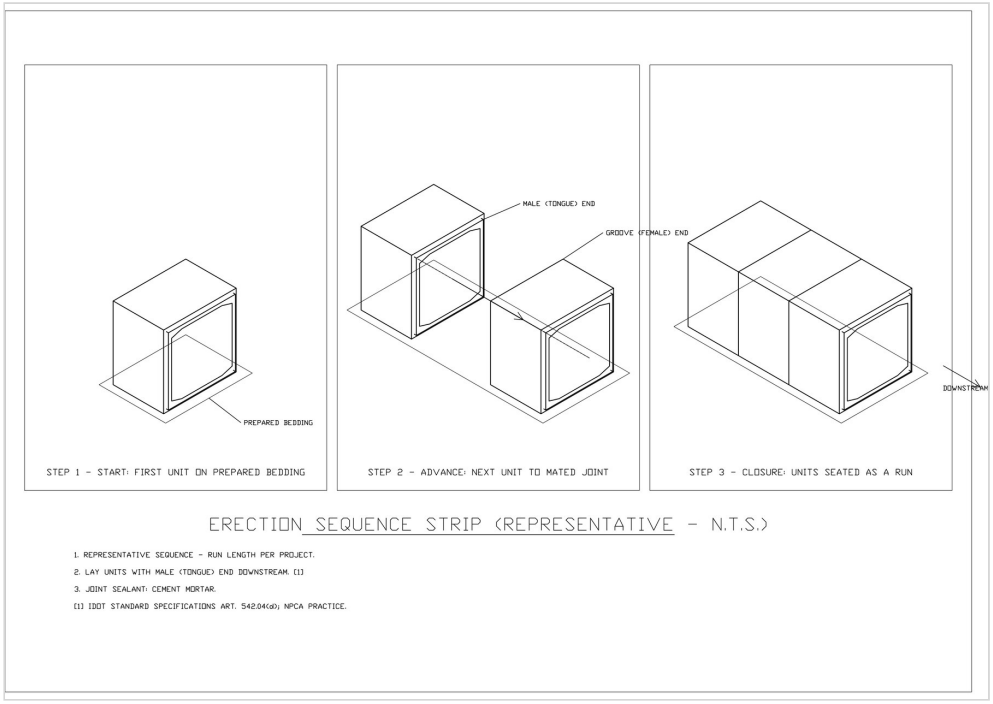


Lifting and rigging isometric with sling angles, lifting data and centre of gravity.

Box culverts: a frame on Winkler soil



Shop drawing: exploded tongue-and-groove joint with sealant and mortar bed.



Erection sequence: first unit on prepared bedding, next unit advanced to the mated joint, closure.

Jacking pipes for microtunnelling

BS EN 1916 and DWA-A 161: ring, jacking force, joints, end zones and fatigue

Jacking pipes are pushed through the ground: besides earth and traffic they carry the jacking force through a deflected joint, and the end zones and collars decide the capacity.

What I built

- Design basis: BS EN 1916 and BS 5911-1 with the DWA-A 161 static calculation, ATV-DVWK-A 127 soil loads and EN 1992-1-1 section checks.
- Declared jacking force as the lower of EN 1916 and DWA-A 161 at the design joint angle, with packer stiffness; ring flexure and crack width, fatigue, end-zone splitting and padding ties, collar checks.
- SAP2000 verification: ring frame, a 3D solid model of the pipe end at the collar, and the steel can of the intermediate jacking station for stress and buckling.
- A range of 21 moulds with two cages, 50 and 100-year design lives, lined and unlined.
- Revit drawing sets per size: general arrangement, joint with an exploded 3D half section, 3D cages and BBS; intermediate jacking station sets for the male and female units.

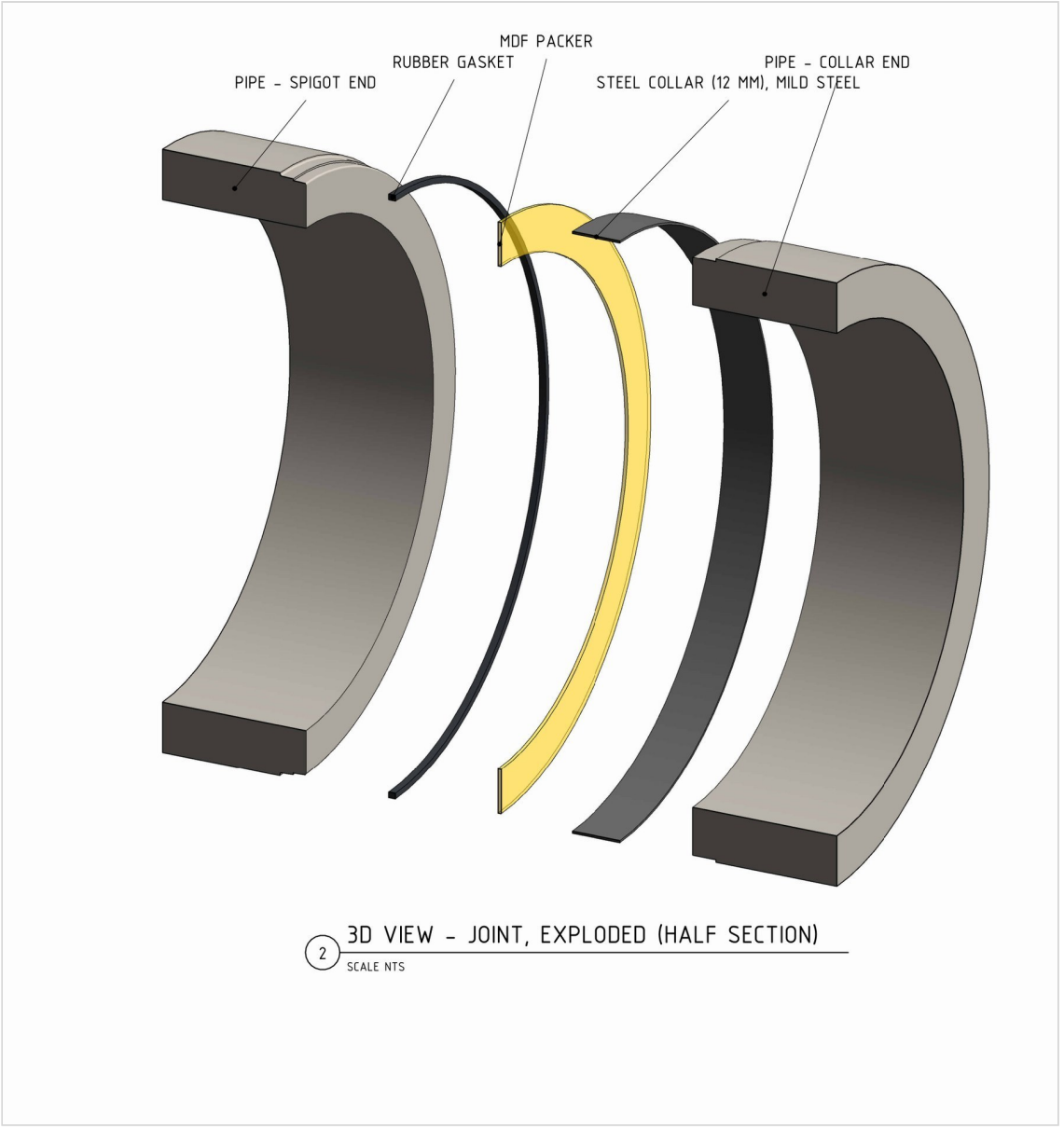
Results

21
moulds in the range

2
cages in every size

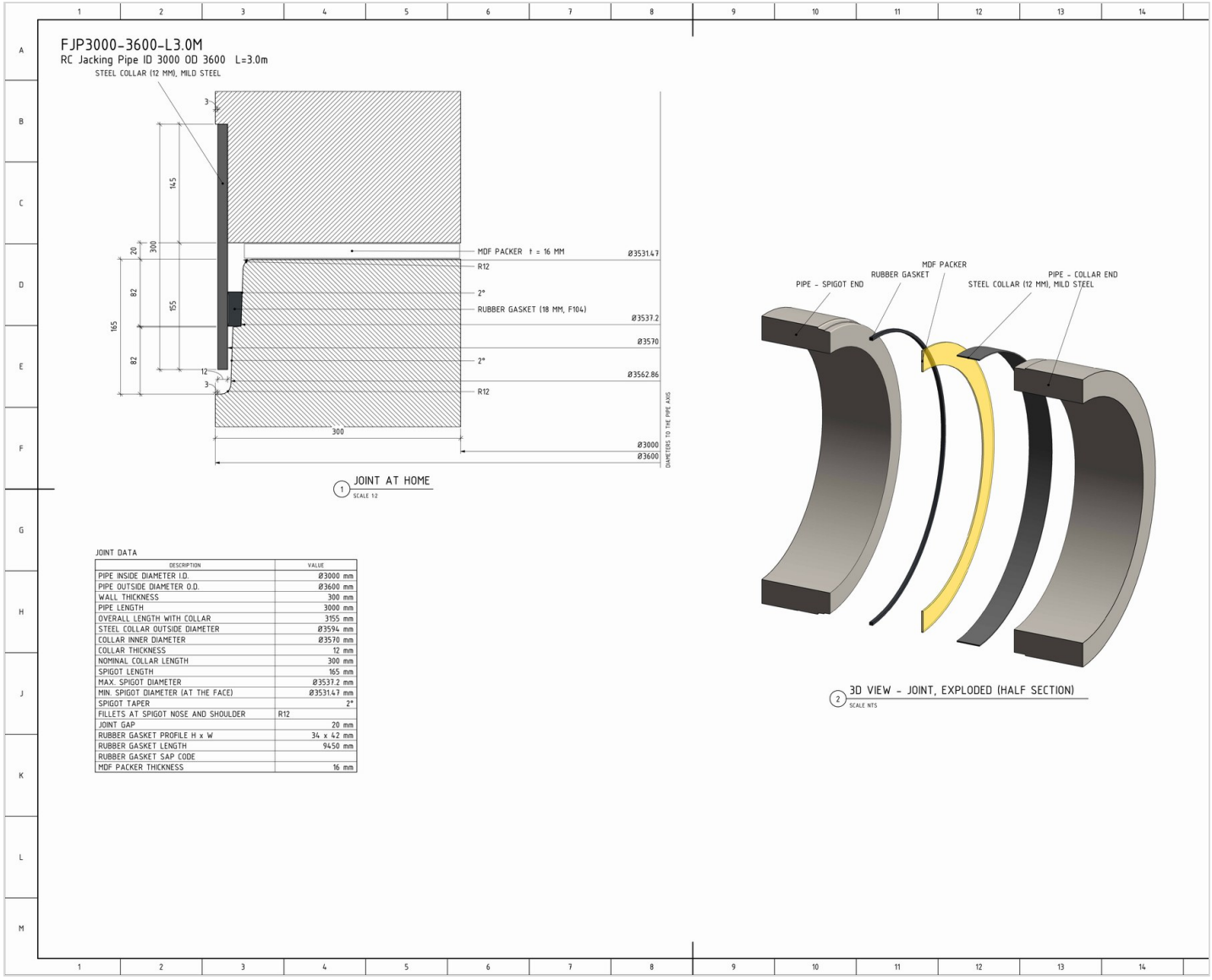
50 / 100
year design lives

Python design engine, SAP2000, Revit (C#), Excel



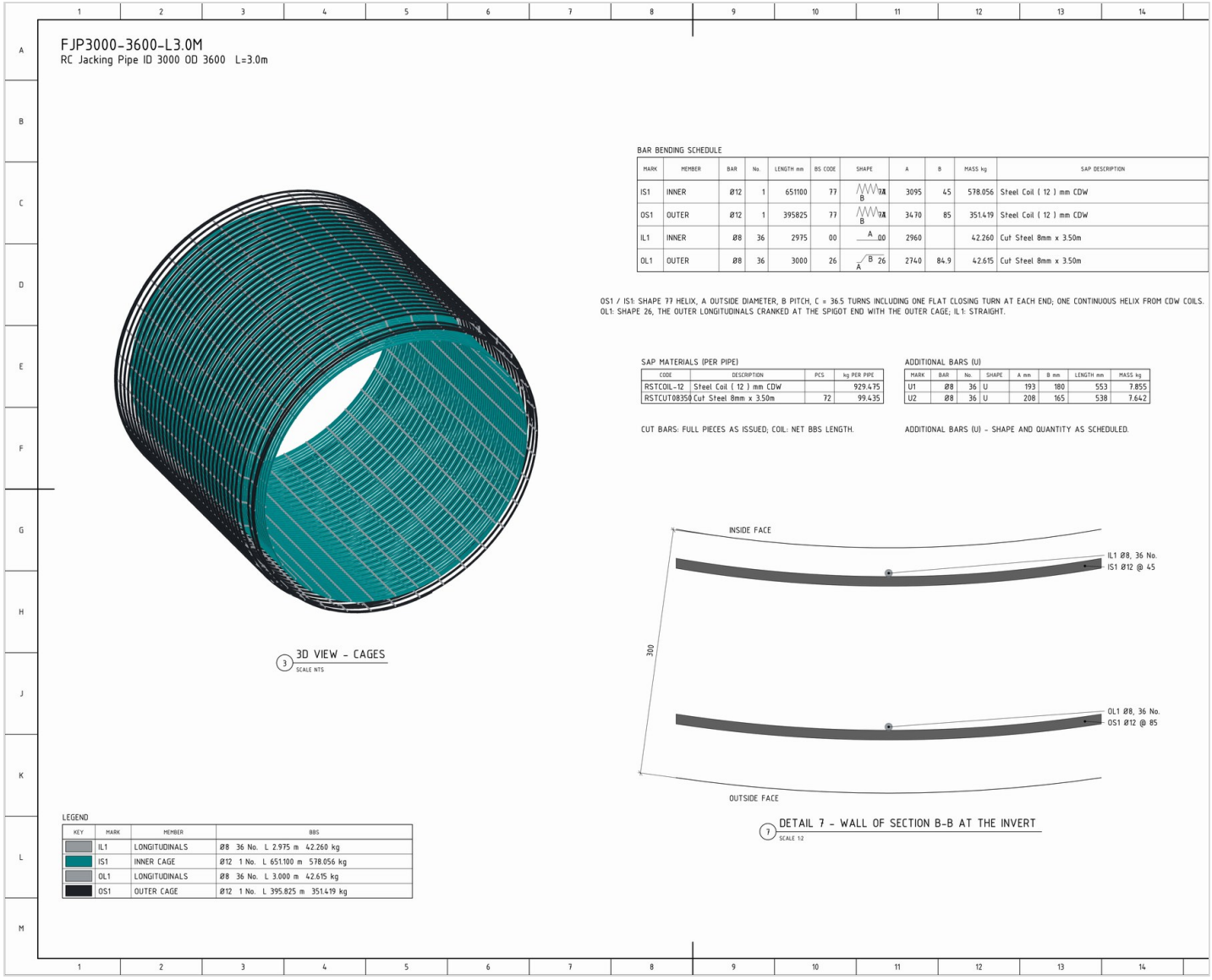
Jacking-pipe joint in Revit: exploded half section of pipe, rubber gasket, packer and steel collar.

Jacking pipes for microtunnelling



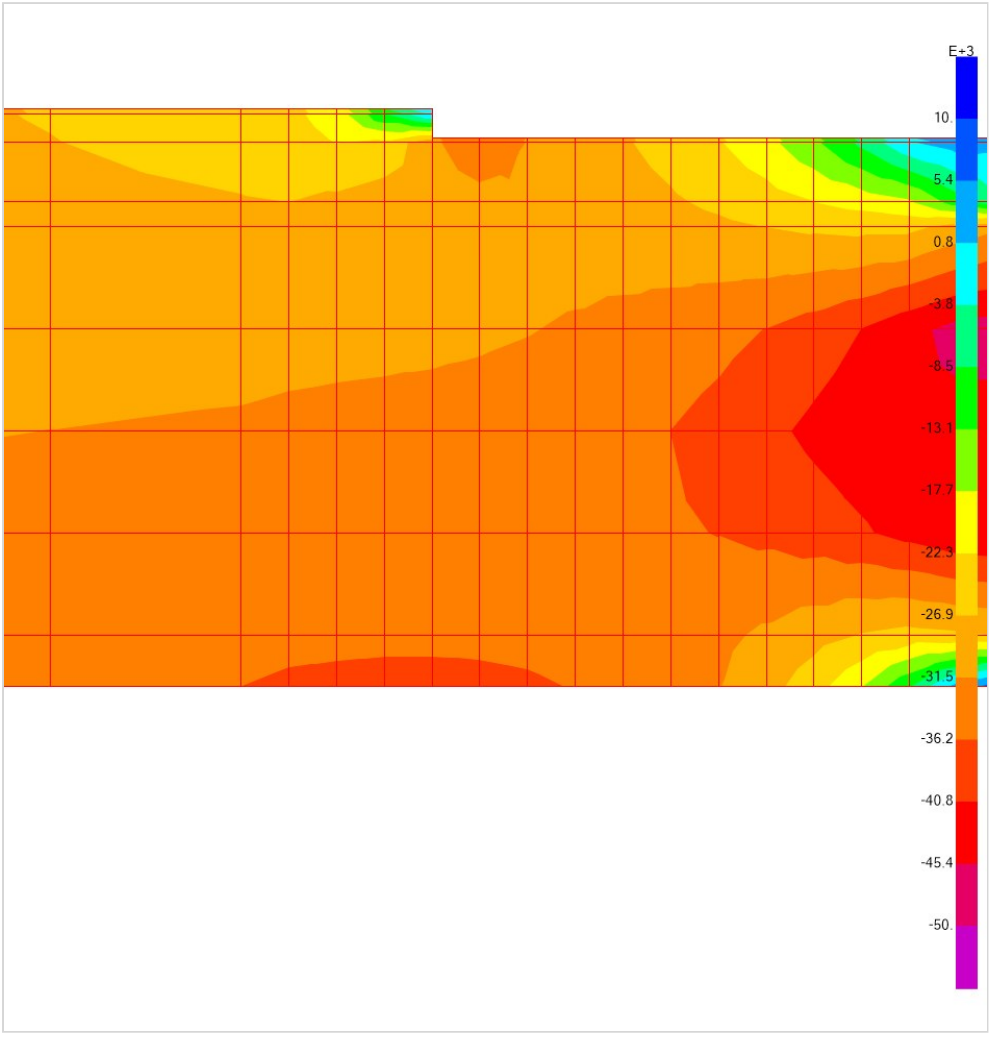
Joint sheet, ID 3000: joint at home with its dimensions, joint data table and the exploded 3D half section.

Jacking pipes for microtunnelling

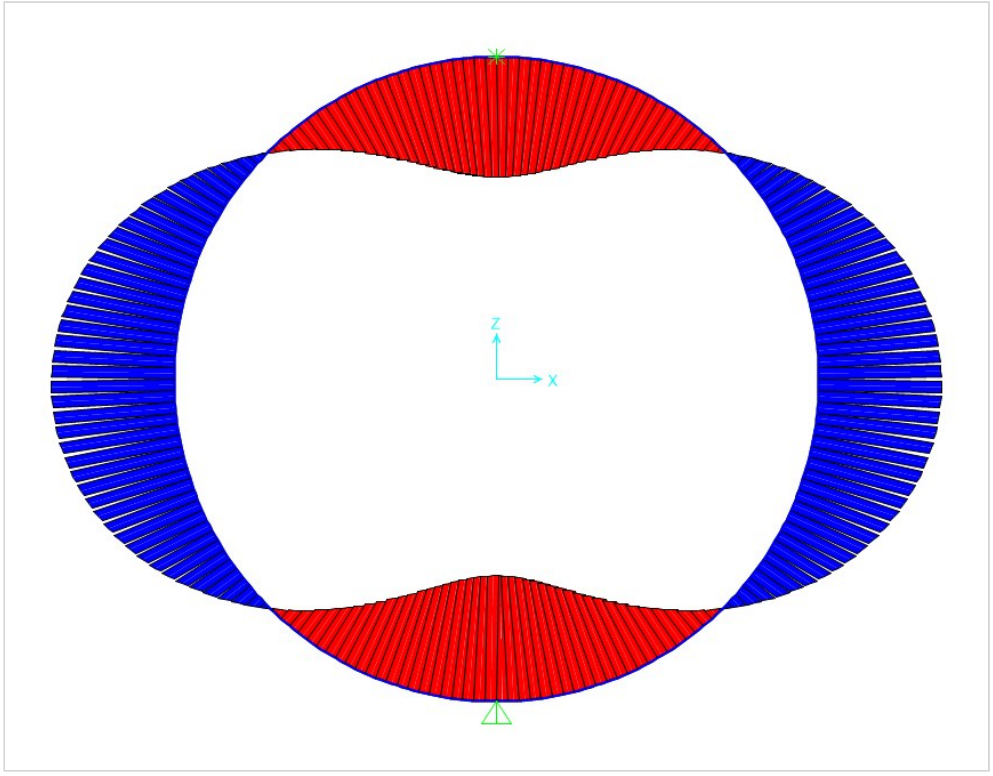


Reinforcement sheet, ID 3000: the two cages in 3D, bar schedule and the wall section at the invert.

Jacking pipes for microtunnelling

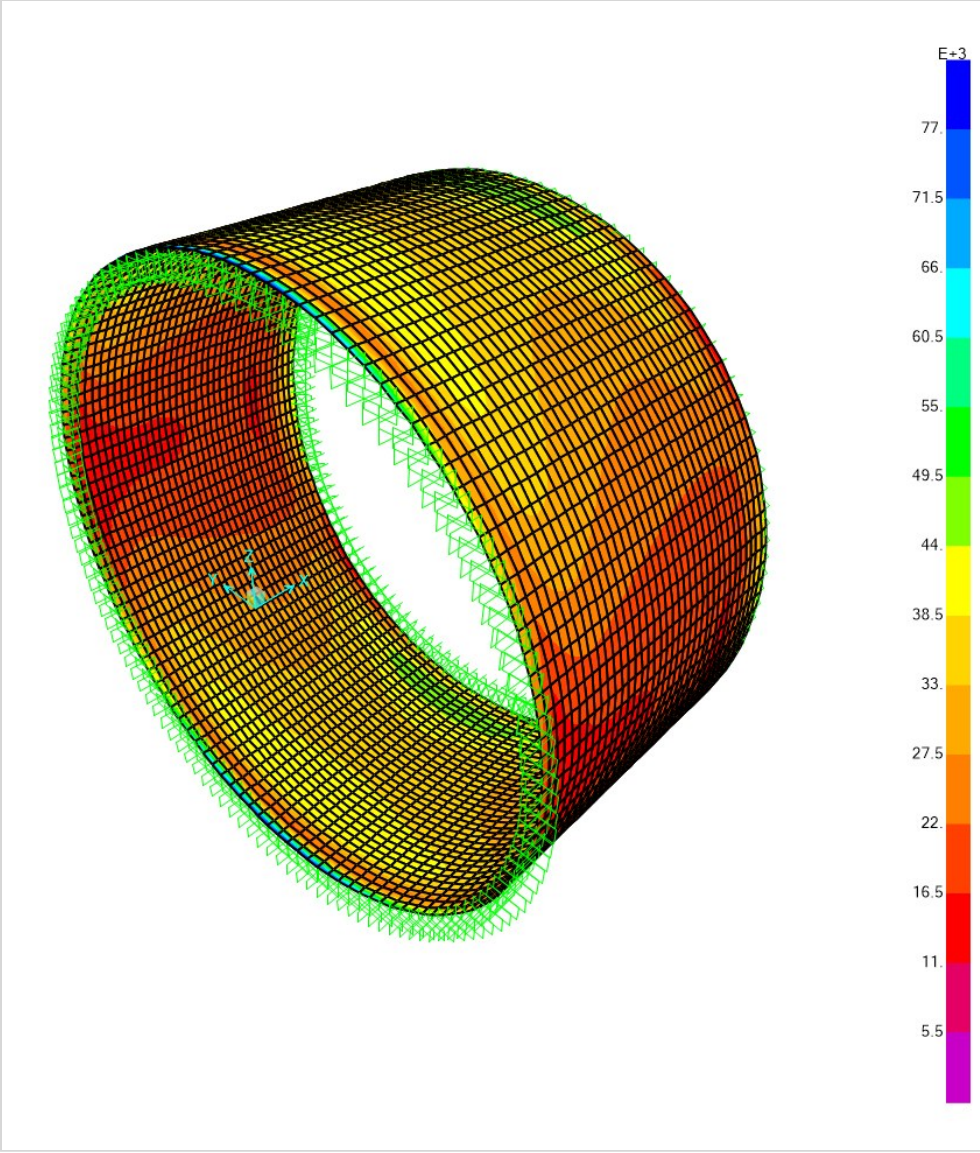


SAP2000 3D solid model of the pipe end: S11 stress at the collar under the jacking force.

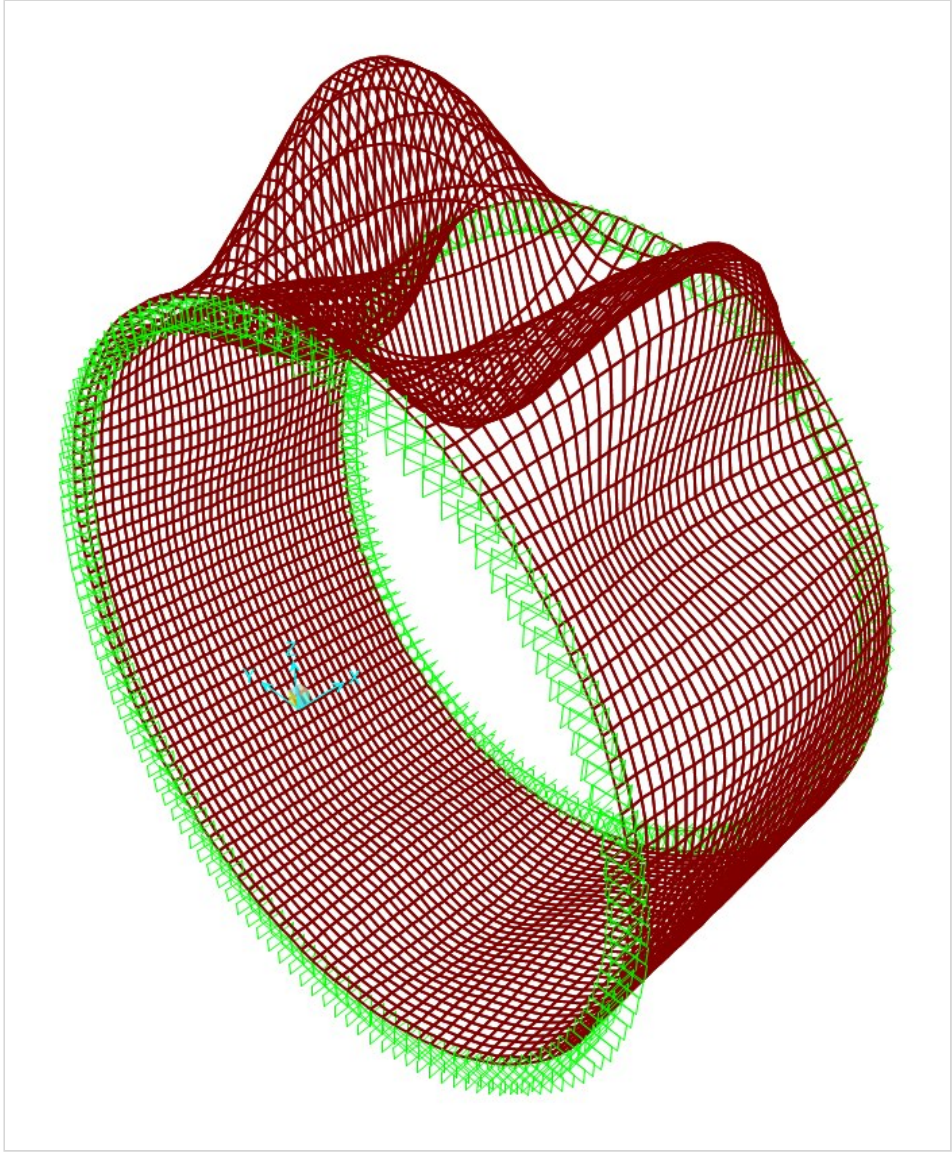


SAP2000 ring frame of the buried pipe: bending moment under earth and traffic loads.

Jacking pipes for microtunnelling



SAP2000 model of the intermediate jacking station steel can: von Mises stress under jacking.



Intermediate jacking station steel can: SAP2000 buckling mode.

Steel moulds for jacking pipes, drawn in BIM

A Revit mould set for every size: parts, casting assemblies, details and design checks

Each pipe size needs its own steel mould: a jacket, a collapsible core, a pallet and a header that cast the joint geometry exactly, carry the fresh-concrete pressure under vibration and hold the product tolerances cast after cast.

What I built

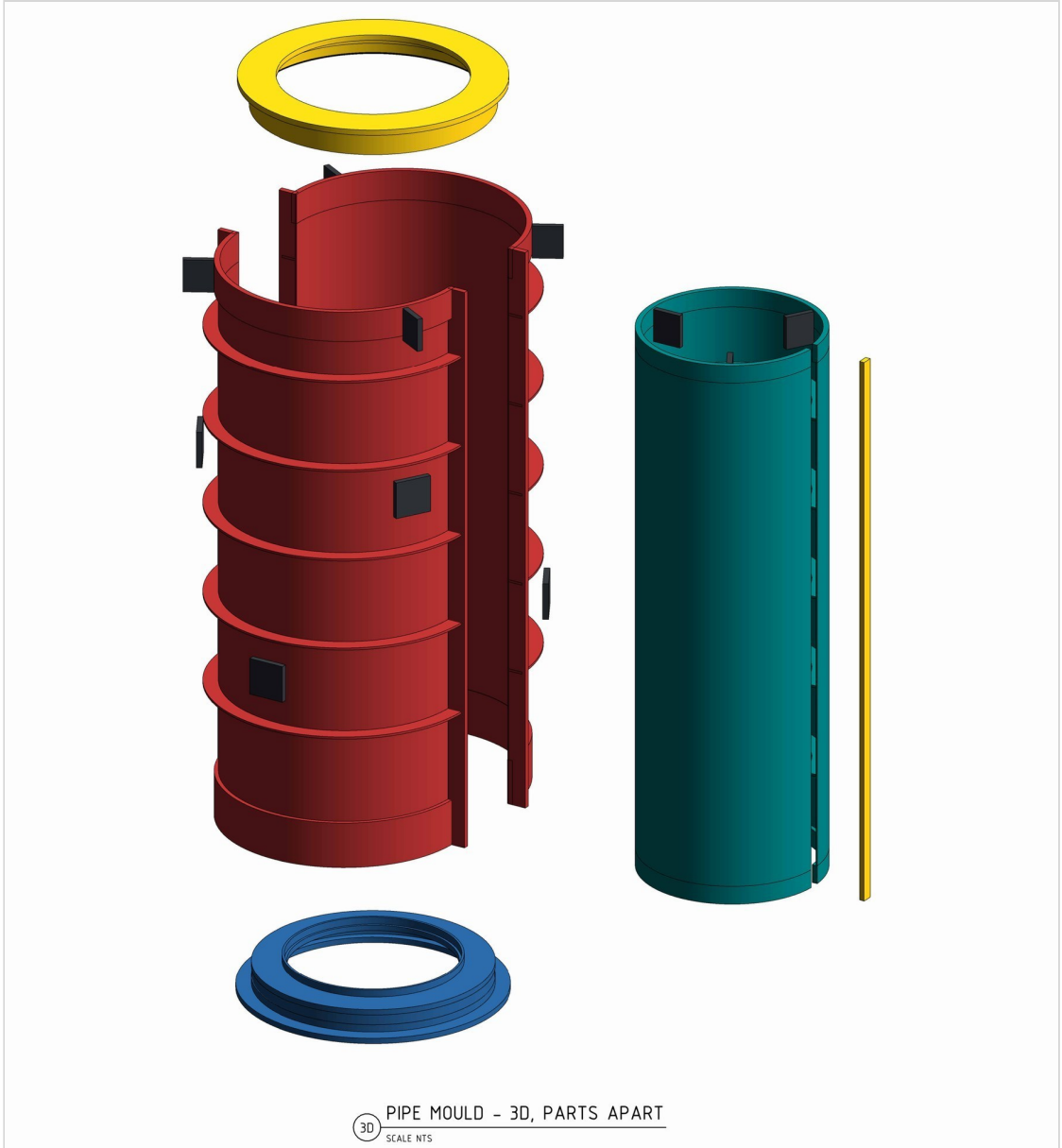
- A mould set generated per size from the same product data as the pipe: cover with the parts in 3D, casting assemblies, pallet, jacket halves, collapsible core, header, intermediate jacking station units and a seal-zone gauge.
- Design checks per part: fresh-concrete pressure as a full liquid head with a factor of 1.5 for vibration, lifting lugs to EN 1993-1-8, and the governing ratio printed for every part.
- Fabrication rules on every set: S275JR steel to BS EN 10025, welding to BS EN ISO 3834-3 and ISO 5817, mould tolerances derived from BS EN 1916 Table G.3, and shop checks before the first cast.
- Automated drawing guards hold every sheet to the drafting standard before it is issued.

Results

21 of 21
mould sets passing the design checks

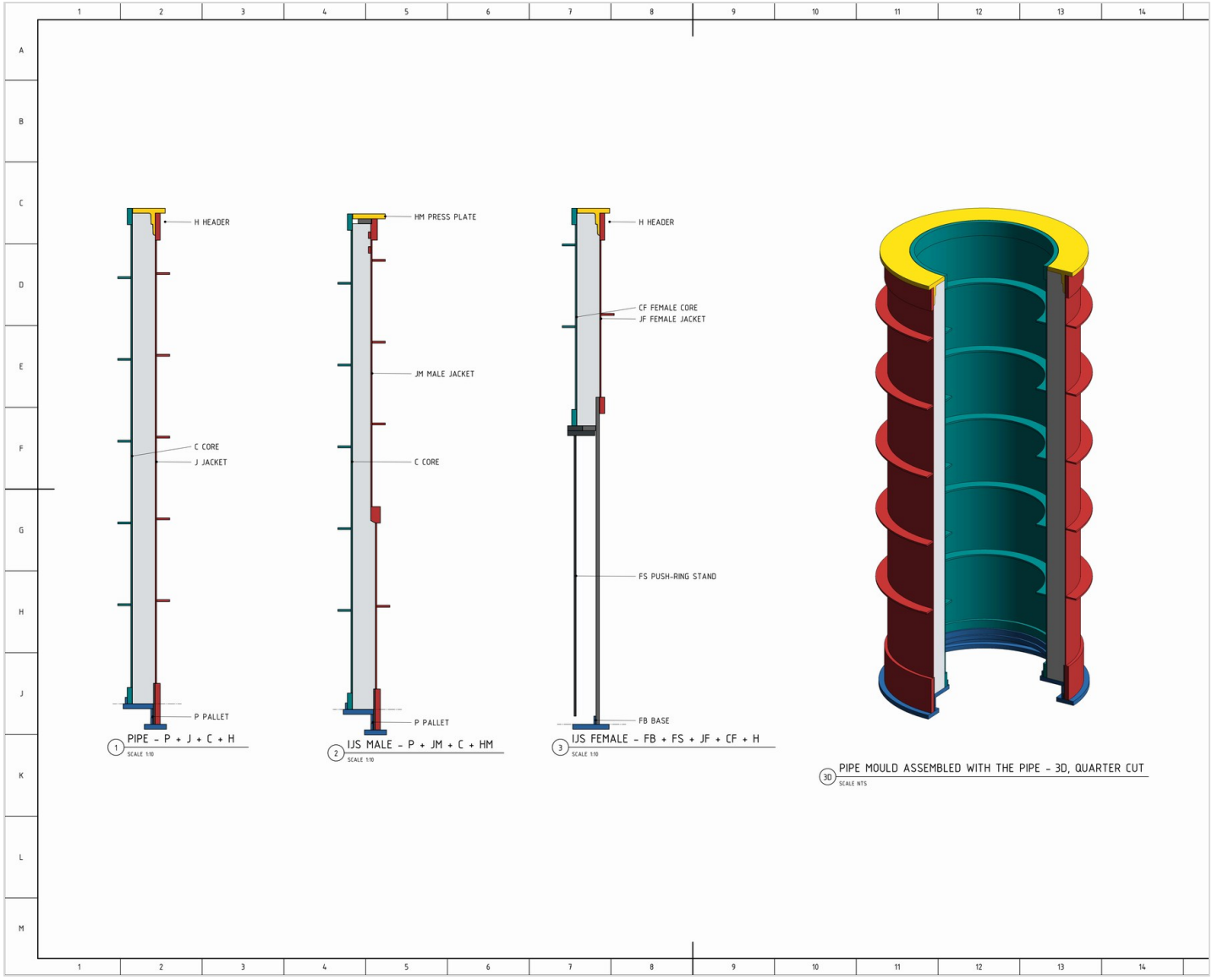
77
automated drawing guards

Revit (C#), Python, Excel



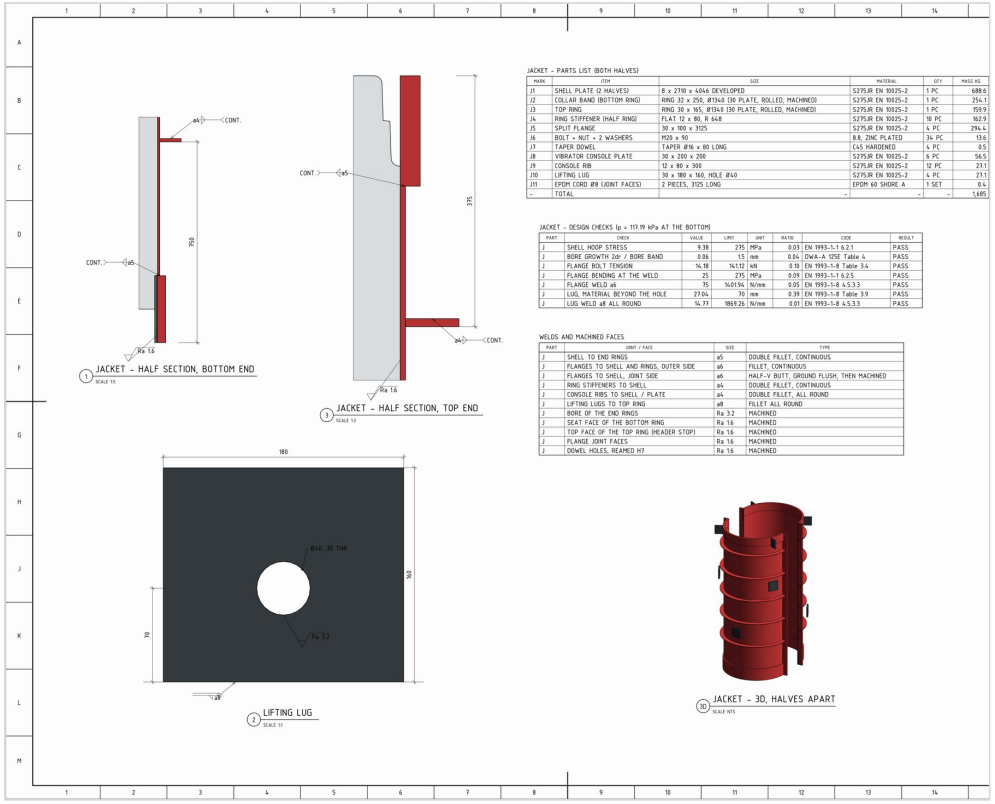
Steel mould for a jacking pipe, parts apart in 3D: header, jacket halves, core and pallet.

Steel moulds for jacking pipes, drawn in BIM

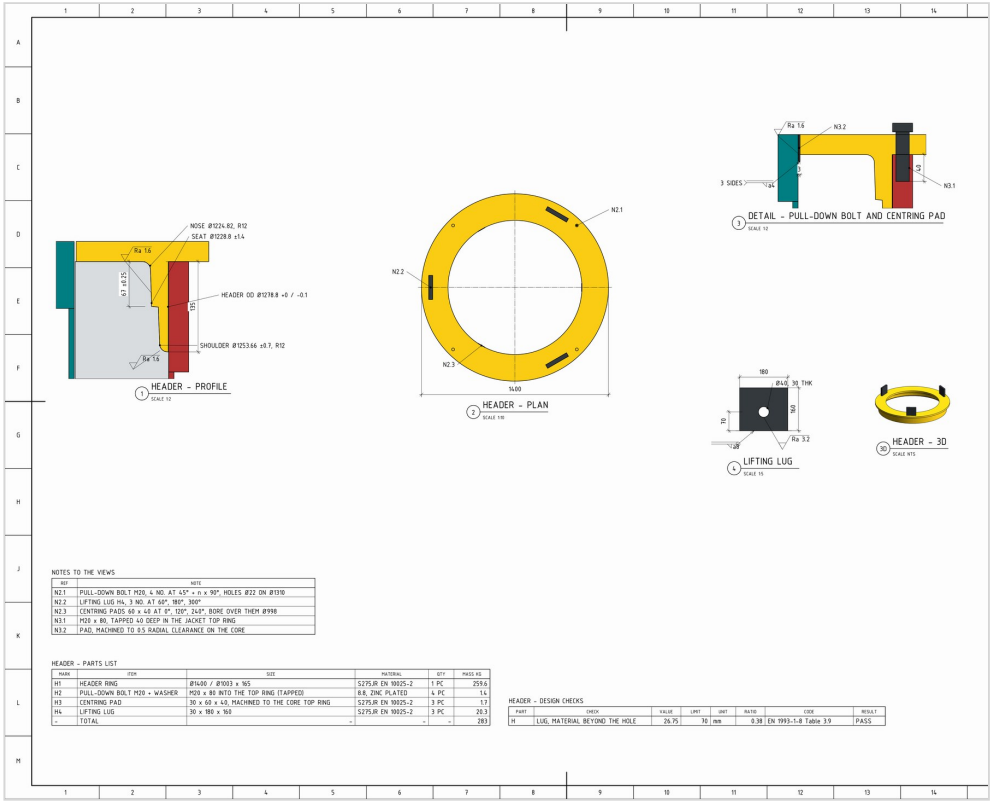


Casting assemblies: pipe, IJS male and IJS female stacks in section, and the assembled mould with the pipe in a quarter cut.

Steel moulds for jacking pipes, drawn in BIM



Jacket details: half sections at the bottom and top ends, parts list, lifting lug and 3D halves.



Header (spigot ring): profile, plan, pull-down bolt and centring pad, lifting lug.

Steel moulds for jacking pipes, drawn in BIM

TOLERANCES (DEVIATION FROM THE DRAWN VALUE, MM)

DIMENSION	PRODUCT	MOULD
OD (BARREL) – JACKET BORE	+0 / −5	−0.5 / −2
ID – CORE	±8	−3 / −6
LENGTH	±8	±2.5
END FACES, ACROSS THE DIAMETER	+3.5 / −0	+1.1 / 0
END FACES, ACROSS THE WALL	+3 / −0	+1 / 0
STRAIGHTNESS	+5 / −0	+1.5 / 0
SPIGOT SEAT – HEADER	+4.4 / −20.8	±1.4
SHOULDER – HEADER	+2.34	±0.7
IJS MALE SEAL ZONE – MALE JACKET	+10 / −21.95	±3.5

DESIGN CHECKS – GOVERNING CHECK PER PART, ALL CHECKS ON THE PART SHEETS

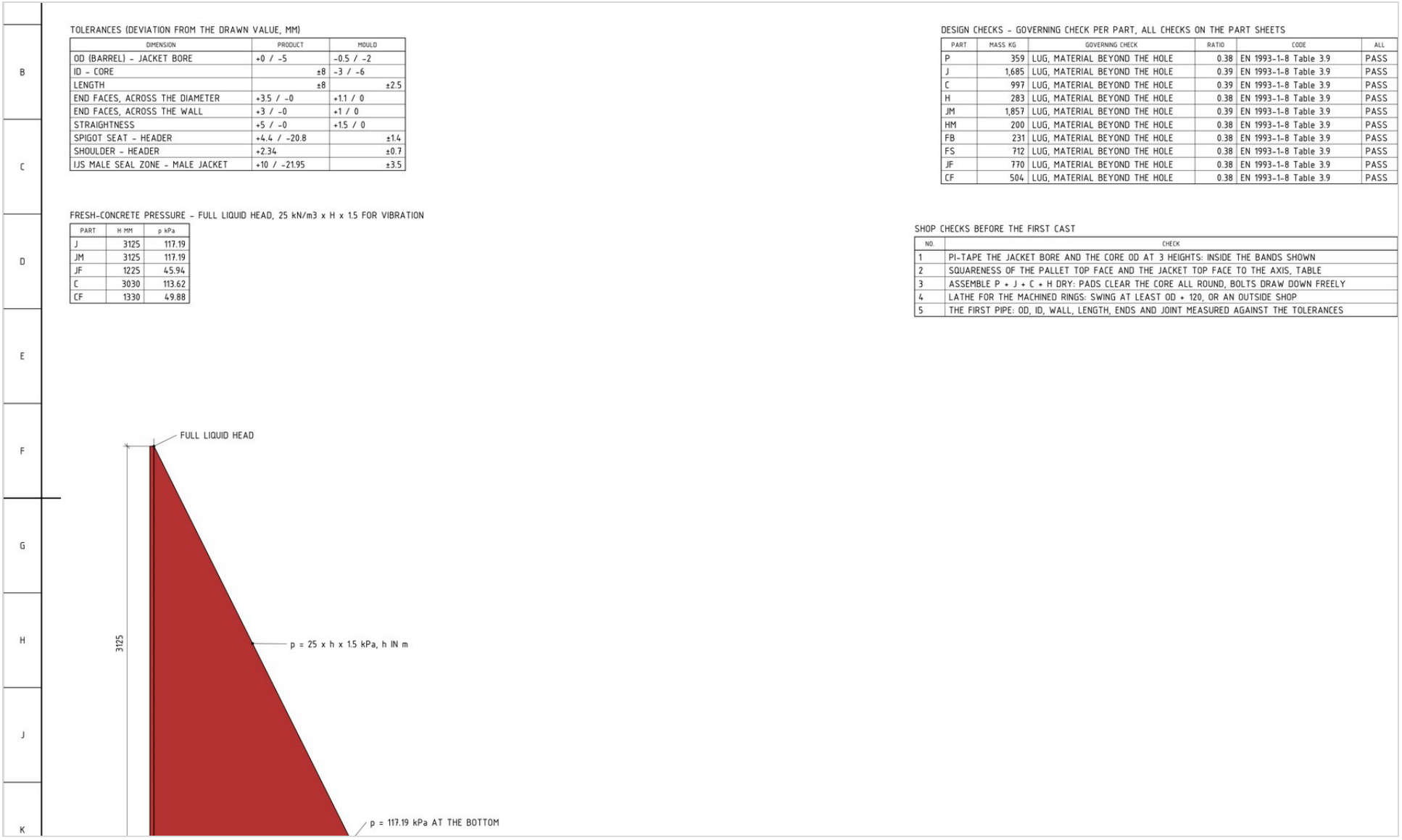
PART	MASS KG	GOVERNING CHECK	RATIO	CODE	ALL
P	359	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
J	1,685	LUG, MATERIAL BEYOND THE HOLE	0.39	EN 1993-1-8 Table 3.9	PASS
C	997	LUG, MATERIAL BEYOND THE HOLE	0.39	EN 1993-1-8 Table 3.9	PASS
H	283	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
JM	1,857	LUG, MATERIAL BEYOND THE HOLE	0.39	EN 1993-1-8 Table 3.9	PASS
HM	200	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
FB	231	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
FS	712	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
JF	770	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS
CF	504	LUG, MATERIAL BEYOND THE HOLE	0.38	EN 1993-1-8 Table 3.9	PASS

FRESH-CONCRETE PRESSURE – FULL LIQUID HEAD, 25 kN/m3 x H x 1.5 FOR VIBRATION

PART	H MM	p kPa
J	3125	117.19
JM	3125	117.19
JF	1225	45.94
C	3030	113.62
CF	1330	49.88

SHOP CHECKS BEFORE THE FIRST CAST

NO.	CHECK
1	PI-TAPE THE JACKET BORE AND THE CORE OD AT 3 HEIGHTS: INSIDE THE BANDS SHOWN
2	SQUARENESS OF THE PALLET TOP FACE AND THE JACKET TOP FACE TO THE AXIS, TABLE
3	ASSEMBLE P + J + C + H DRY: PADS CLEAR THE CORE ALL ROUND, BOLTS DRAW DOWN FREELY
4	LATHE FOR THE MACHINED RINGS: SWING AT LEAST OD + 120, OR AN OUTSIDE SHOP
5	THE FIRST PIPE: OD, ID, WALL, LENGTH, ENDS AND JOINT MEASURED AGAINST THE TOLERANCES



Tolerances from BS EN 1916, governing design check per part to EN 1993-1-8, fresh-concrete pressure on the jacket and shop checks before the first cast.

Approvals, submittals and technical documents

Memos, catalogues, drawings and prequalification packages prepared for technical review

Consultants approve a precast supplier on paper first: prequalification, material submittals, calculation memos, drawings and test plans, each in the client's format and each needing current certificates.

What I built

- Calculation memos generated by the design engines, every check traced to its source: 156 RC pipe memos (76 direct, 80 indirect) and full manhole and box culvert memos with SAP2000 read-back.
- Technical catalogues for the RC pipe range: an 86-page engineering catalogue and a 35-page bilingual public catalogue.
- A prequalification and material-submittal system: an 18-category reference library, a certificate-expiry register, compliance sheets per product and a folder per client request.
- The master prequalification package rebuilt in the company identity: 452 pages, 156 bookmarks and a section file per chapter.
- Inspection and test plans, including three-edge-bearing proof testing to ASTM C497.

Results

156
RC pipe design memos

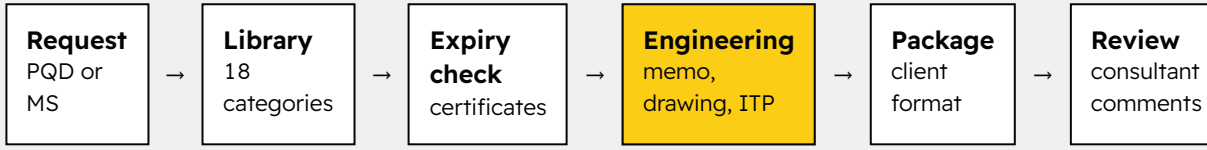
452
page prequalification package

18
category reference library

86
page technical catalogue

Python memo and catalogue generators, Word and PDF, Revit, Excel

From a technical request to a review-ready submission



Documents I issue

Memos	RC pipes (direct and indirect), manholes, box culverts, jacking pipes
Catalogues	Engineering and bilingual public catalogues for the RC pipe range
Drawings	Standard Revit sets with BBS for every product family and the steel moulds
Quality	Inspection and test plans and compliance sheets, packed with mix designs and certificates
Packages	Prequalification and material submittals in the client's format

Tender analysis and pricing

~4,000 quotations since 2023 across Saudi giga- and mega-projects

Client BOQs arrive in Arabic and English with mixed conventions, and every line has to become a product the factory can design, price, make and supply.

What I built

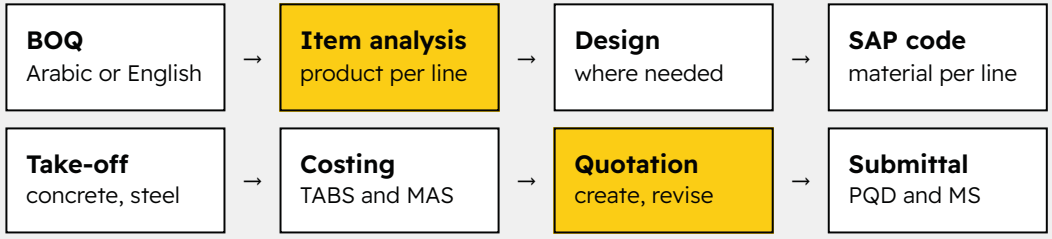
- BOQ and item analysis, product design where needed, quantity take-off, costing and drawings for every quotation; SAP material creation since Go-Live.
- A gated take-off pipeline that resolves BOQ lines to SAP material codes and turns every error found into a reusable check.
- TABS activity-based costing used with MAS, and a pricing engine with floor, target and max per family.
- Prequalification and material submittals backed by an 18-category reference library and a certificate-expiry register.

Results

~4,000
quotations since 2023

Python, Excel, OCR and PDF parsing, SAP S/4HANA

From the client's BOQ to a priced, approvable offer



Every quotation carries its engineering: the design check, the drawing and the SAP product it will be made as.

Tender analysis and pricing

Table 1: Manholes — grouped by internal diameter, then by depth tier (H≤3M · 3<H≤6M · H>6M). Item count and Σ Qty shown per tier.

No.	Material Type	Material Group	SAP Code	SAP Description	Description	ID / OD (mm)	Height (mm)	Wall (mm)	Base (mm)	Thickness (mm)	Qty	Unit	Remarks
ID 1200 MM													
	H≤3M — 2 item(s) · Σ Qty = 1003												
1	Manhole	Base	FB120X25/30X120A	MANHOLE BASE 1200X250/300X1200 MM H≤3M	توريد مناهل مقاس (١٢٠٠*٣٠٠/٢٥٠*١٢٠٠)مم <٣م	1200	1200	250	300		1002	عدد	Thickness pair: the BOQ writes 300/250 (larger/smaller) = base 300 / wall 250 m
2	Manhole	Base	FB120X25/30X150A	MANHOLE BASE 1200X250/300X1500 MM H≤3M	توريد مناهل مقاس (١٥٠٠*٣٠٠/٢٥٠*١٢٠٠)مم <٣م	1200	1500	250	300		1	عدد	Thickness pair: the BOQ writes 300/250 (larger/smaller) = base 300 / wall 250 m
	3<H≤6M — 1 item(s) · Σ Qty = 496												
3	Manhole	Base	FB120X30/30X120B	MANHOLE BASE 1200X300/300X1200 MM 3<H≤6M	توريد مناهل مقاس (١٢٠٠*٣٠٠/٣٠٠*١٢٠٠)مم ٣-٦م	1200	1200	300	300		496	عدد	Thickness pair: the BOQ writes 300/300 (larger/smaller) = base 300 / wall 300 m
	H>6M — 1 item(s) · Σ Qty = 39												
4	Manhole	Base	FB120X30/30X120C	MANHOLE BASE 1200X300/300X1200 MM H>6M	توريد مناهل مقاس (١٢٠٠*٣٠٠/٣٠٠*١٢٠٠)مم <٦م	1200	1200	300	300		39	عدد	Thickness pair: the BOQ writes 300/300 (larger/smaller) = base 300 / wall 300 m
	Rings — 4 item(s) · Σ Qty = 4												
5	Manhole	Rings	FR120X20X120	MANHOLE RING 1200X200X1200 MM	توريد رنجات مقاس (٢٠٠*١٢٠٠*١٢٠٠)مم	1200	1200	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
6	Manhole	Rings	FR120X20X90	MANHOLE RING 1200X200X900 MM	توريد رنجات مقاس (٢٠٠*٩٠٠*١٢٠٠)مم	1200	900	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
7	Manhole	Rings	FR120X20X60	MANHOLE RING 1200X200X600 MM	توريد رنجات مقاس (٢٠٠*٦٠٠*١٢٠٠)مم	1200	600	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
8	Manhole	Rings	FR120X20X30	MANHOLE RING 1200X200X300 MM	توريد رنجات مقاس (٢٠٠*٣٠٠*١٢٠٠)مم	1200	300	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
	Flat Cover — 1 item(s) · Σ Qty = 1537												
9	Manhole	Flat cover	FFC160/60X25	FLAT COVER 1600/600X250 MM	توريد اغطية مقاس ١٦٠٠*٣٢٥مم	1600				250	1537	عدد	Cover for the ID 1200 manholes: OD 1600 = 1200 + 2 x 200 (ring / chamber wal
ID 1500 MM													
	3<H≤6M — 1 item(s) · Σ Qty = 16												
10	Manhole	Base	FB150X30/35X150B	MANHOLE BASE 1500X300/350X1500 MM 3<H≤6M	توريد مناهل مقاس (١٥٠٠*٣٥٠/٣٠٠*١٥٠٠)مم ٣-٦م	1500	1500	300	350		16	عدد	Thickness pair: the BOQ writes 350/300 (larger/smaller) = base 350 / wall 300 m
	H>6M — 1 item(s) · Σ Qty = 4												
11	Manhole	Base	FB150X30/35X150C	MANHOLE BASE 1500X300/350X1500 MM H>6M	توريد مناهل مقاس (١٥٠٠*٣٥٠/٣٠٠*١٥٠٠)مم <٦م	1500	1500	300	350		4	عدد	Thickness pair: the BOQ writes 350/300 (larger/smaller) = base 350 / wall 300 m
	Rings — 4 item(s) · Σ Qty = 4												
12	Manhole	Rings	FR150X20X120	MANHOLE RING 1500X200X1200 MM	توريد رنجات مقاس (٢٠٠*١٢٠٠*١٥٠٠)مم	1500	1200	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
13	Manhole	Rings	FR150X20X90	MANHOLE RING 1500X200X900 MM	توريد رنجات مقاس (٢٠٠*٩٠٠*١٥٠٠)مم	1500	900	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
14	Manhole	Rings	FR150X20X60	MANHOLE RING 1500X200X600 MM	توريد رنجات مقاس (٢٠٠*٦٠٠*١٥٠٠)مم	1500	600	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
15	Manhole	Rings	FR150X20X30	MANHOLE RING 1500X200X300 MM	توريد رنجات مقاس (٢٠٠*٣٠٠*١٥٠٠)مم	1500	300	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
	Flat Cover — 1 item(s) · Σ Qty = 20												
16	Manhole	Flat cover	FFC190/60X25	FLAT COVER 1900/600X250 MM	توريد اغطية مقاس ١٩٠٠*٣٢٥مم	1900				250	20	عدد	Cover for the ID 1500 manholes: OD 1900 = 1500 + 2 x 200 (ring / chamber wal
ID 2500 MM													
	3<H≤6M — 2 item(s) · Σ Qty = 11												
17	Manhole	Base	FB250X30/35X220B	MANHOLE BASE 2500X300/350X2200 MM 3<H≤6M	توريد مناهل مقاس (٢٢٠٠*٣٥٠/٣٠٠*٢٥٠٠)مم	2500	2200	300	350		10	عدد	Thickness pair: the BOQ writes 350/300 (larger/smaller) = base 350 / wall 300 m
18	Manhole	Base	FB250X30/35X220B	MANHOLE BASE 2500X300/350X2200 MM 3<H≤6M	توريد مناهل مقاس (٢٢٠٠*٣٥٠/٣٠٠*٢٥٠٠)مم ٤-٦م	2500	2200	300	350		1	عدد	Thickness pair: the BOQ writes 350/300 (larger/smaller) = base 350 / wall 300 m
	Rings — 4 item(s) · Σ Qty = 4												
19	Manhole	Rings	FR250X20X120	MANHOLE RING 2500X200X1200 MM	توريد رنجات مقاس (٢٠٠*١٢٠٠*٢٥٠٠)مم	2500	1200	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
20	Manhole	Rings	FR250X20X90	MANHOLE RING 2500X200X900 MM	توريد رنجات مقاس (٢٠٠*٩٠٠*٢٥٠٠)مم	2500	900	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
21	Manhole	Rings	FR250X20X60	MANHOLE RING 2500X200X600 MM	توريد رنجات مقاس (٢٠٠*٦٠٠*٢٥٠٠)مم	2500	600	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
22	Manhole	Rings	FR250X20X30	MANHOLE RING 2500X200X300 MM	توريد رنجات مقاس (٢٠٠*٣٠٠*٢٥٠٠)مم	2500	300	200			1	عدد	Ring string read as ID * height * wall (wall 200 closes onto the cover OD: ID + 2
	Flat Cover — 1 item(s) · Σ Qty = 11												

Tender take-off excerpt: manhole lines of the client's Arabic BOQ resolved to SAP material codes with dimensions and quantities.

Precast Engineering ERP

One inquiry record from the client's BOQ to a production-ready SAP product

Every order needed its own take-off, drawings, memos and submittals in separate files, and SAP products were created one screen at a time, so the price, the drawing and the SAP product could drift apart.

What I built

- Inquiry-centred platform: BOQ import, item analysis, cost take-off from live SAP BOMs, quotation create and revise, new FG/BOM/routing/QM creation, with the design, drawing and submittal engines linked into the same record.
- Every tool follows one contract: a ribbon button, a JSON job, then preview, apply, verify and log. SAP writes stay locked until each operation is approved.
- Version 1 runs daily; version 2 rebuilds it on a tested Python core with 4,252 automated tests.

Results

1,290+
inquiries run through it since Go-Live

4,252
tests on the version 2 core

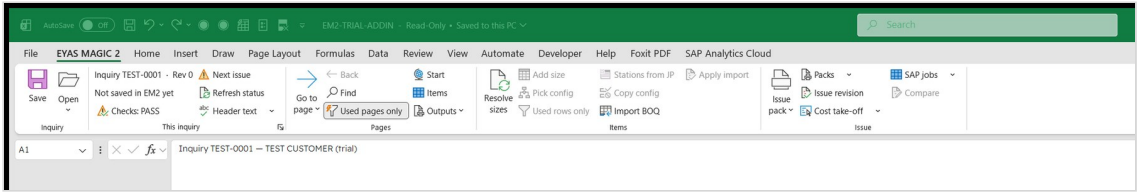
Excel ribbon add-in, VBA, Python, SQLite, SAP S/4HANA OData

One inquiry record runs the whole order

Inquiry client BOQ → **Item analysis** SAP codes → **Cost take-off** live SAP BOMs → **Quotation** create, revise → **SAP product** FG, BOM, routing

Every tool follows the same contract

Ribbon one button → **Job** JSON in and out → **Preview** before any write → **Apply** approved only → **Verify** read back → **Log** audit trail



The ERP's Excel ribbon (test inquiry).

Precast Engineering ERP

Request for quotation

Line items: material codes, mix designs, coatings and quantities

Sales Inquiry	TEST-0001	Revision	1
Sales Quotation	—	Date	17/10/2026
Customer	TEST CUSTOMER (trial)	Contact Person	ENG. E2E
Project	Trial of EYAS MAGIC 2 — item entry	Contact No.	—
Site Location	JEDDAH	JDCO Sales Person	TEST

1 Line items

No.	Material Code	Description	Mix Design	Cage Type	External Coating	Internal Coating	Qty	Unit	Remarks
1	FJP1000-1280-L3.0M	RC Jacking Pipe ID 1000 OD 1280 L=3.0m	MD015-SRC-41.4	STANDARD	300 MIC BITUMEN	UN-COATED	9	m	FC:12m -WT:NA -JFC:420 TON
2	FIJS-F-1000-1280	IJS Female ID 1000 OD 1280	MD015-SRC-41.4	STANDARD	300 MIC BITUMEN	UN-COATED	4	PC	FC:12m -WT:NA -JFC:420 TON
3	FIJS-F-1200-1490	IJS Female ID 1200 OD 1490	MD-TEST-JP	STANDARD	300 MIC BITUMEN	UN-COATED	1	PC	FC:12m -WT:NA -JFC:500 TON
4	FIJS-M-1200-1490	IJS Male ID 1200 OD 1490	MD-TEST-JP	STANDARD	300 MIC BITUMEN	UN-COATED	1	PC	FC:12m -WT:NA -JFC:500 TON
5	FR120X18X30	MANHOLE RING 1200X180X300 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
6	FR120X18X60	MANHOLE RING 1200X180X600 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
7	FR120X18X90	MANHOLE RING 1200X180X900 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
8	FR120X18X120	MANHOLE RING 1200X180X1200 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
9	FEC120/60X65W18	Eccentric Cone 1200/600X650 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
10	FEC120/60X100W18	Eccentric Cone 1200/600X1000 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	2	PC	
11	FFC156/60X25	FLAT COVER 1560/600X250 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	5	PC	
12	FR150X25X30	MANHOLE RING 1500X250X300 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
13	FR150X20X60	MANHOLE RING 1500X200X600 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
14	FR150X20X90	MANHOLE RING 1500X200X900 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
15	FR150X20X120	MANHOLE RING 1500X200X1200 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
16	FTS190/120X30	CR.T.SLAB 1900/1200X300 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
17	FR200X20X30	MANHOLE RING 2000X200X300 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	3	PC	
18	FR200X20X60	MANHOLE RING 2000X200X600 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
19	FR200X20X90	MANHOLE RING 2000X200X900 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
20	FR200X20X120	MANHOLE RING 2000X200X1200 MM	MD019-SRC-35	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
21	FFC240/60X25	FLAT COVER 2400/600X250 MM	MD-E2E-COVER	STANDARD	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	5	PC	
22	FR90X20X30	MANHOLE RING 900X200X300 MM	MD019-SRC-35	STANDARD	600 MIC BITUMEN	1000 MIC COAL TAR EPOXY	2	PC	
23	FFC130/60X20	FLAT COVER 1300/600X200 MM	MD019-SRC-35	STANDARD	600 MIC BITUMEN	1000 MIC COAL TAR EPOXY	3	PC	
24	FSR808030W20	SQ. Ring 800x800x300 W200 MM	MD019-SRC-35	12@150	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	
25	FSR808060W20	SQ. Ring 800x800x600 W200 MM	MD019-SRC-35	12@150	450 MIC BITUMEN	400 MIC COAL TAR EPOXY	1	PC	

Generated request-for-quotation pack: every BOQ line resolved to a SAP material code, mix design and coating (test data).

Precast Engineering ERP

Concrete mix design take-off

Concrete mix take-off per variant

Sales Inquiry	**TEST-0001**	Revision	**1**
Customer	**TEST CUSTOMER (trial)**	Sales Quotation	**—**
Project	**Trial of EYAS MAGIC 2 — item entry**	Location	**JEDDAH**

1 Mix design legend

Table 4: Mix Design codes used in this RFQ.

No.	Code	Description
1	MD015-SRC-41.4	SRC CEMENT TYPE V - 41.4 MPA
2	MD-TEST-JP	
3	MD019-SRC-35	SRC CEMENT TYPE V - 35 MPA
4	MD-E2E-COVER	

2 Concrete mix design take-off

Table 3: concrete mix take-off per variant.

No.	Description	QTY	Unit	Vol (m^3)	Conc. Wt	Mix Design	SRC kg	Agg 1/2" kg	Agg 3/8" kg	White sand kg	Water L	Structuro 505 L
1	RC Jacking Pipe ID 1000 OD 1280 L=3.0m	1	PC	1.50	3.76	MD015-SRC-41.4	676.80	979.10	800.13	870.82	209.06	10.53

Concrete-mix take-off computed per item from the SAP BOMs: cement, aggregates, sand, water (test data).

SAP S/4HANA Cloud: master data to Go-Live

A manufacturing product model built from an empty system

JDCO moved onto SAP S/4HANA Cloud (Public Edition) with no prior system data. Engineering, production, planning, sales and finance needed one product model they could all trust from day one.

What I built

- Material coding standard, BOM and routing logic, work centers linked to cost centers, and classification/AVC for configurable box culverts.
- Excel/VBA and Python generators that produce FG, BOM, routing and inspection data from reference products, loaded through Migration Cockpit and OData.
- Engineering connected to costing, procurement, production and planning through one product model.
- Training for the technical office and users across departments, then first-line support after Go-Live.

Results

~40,000
materials across 11 material types

34,000+
BOMs, ~1.2M items

~11,000
routings, 41,000+ operations

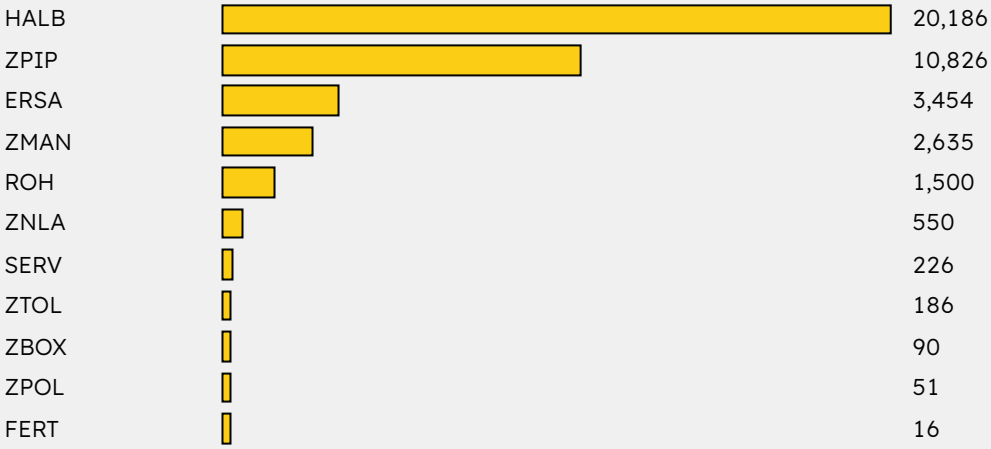
30 Dec 2025
Go-Live

SAP S/4HANA Cloud MM/PP, Migration Cockpit, Mass Maintenance, OData, Python, VBA

One product model for every department



Material master by type, April 2026 extract



SAP extract, April 2026.

Analytics and planning on S/4HANA

Dashboards and planning tools from live SAP data

Management needed production, dispatch and material-planning views straight from SAP, without manual exports.

What I built

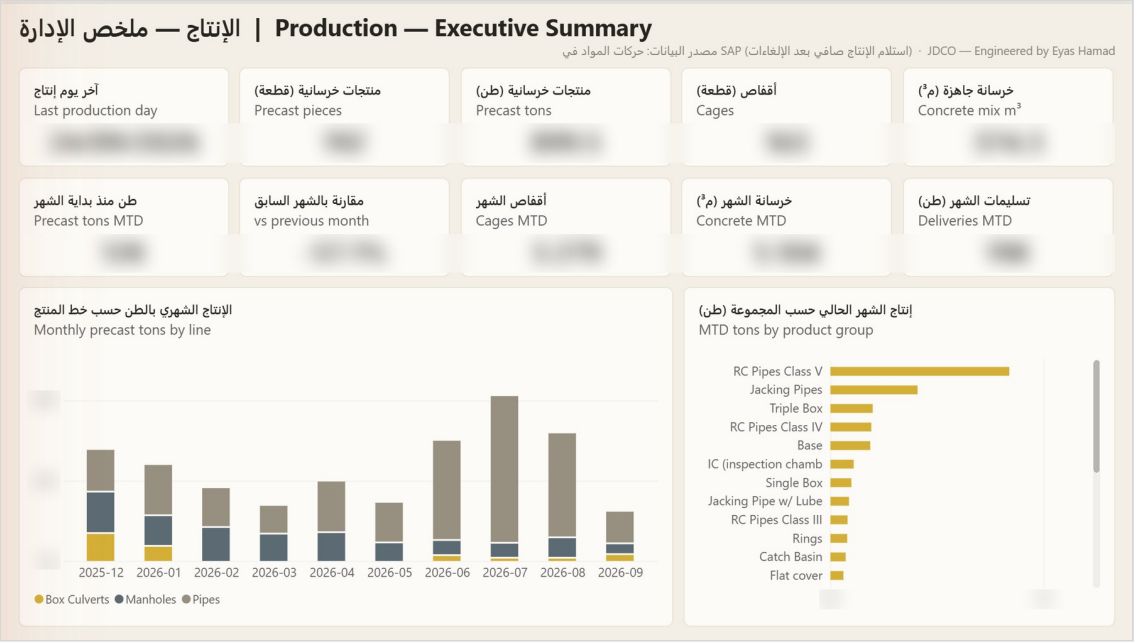
- A 35-entity OData extraction and local data mirror (~9M rows) with a data dictionary and join map.
- 29 scheduled HTML and Excel reports, and a code-generated Power BI production model.
- A raw-material plan that checks the production plan against SAP stock and open purchase orders.

Results

29
automated management reports

~9M
rows mirrored from SAP

Python, SAP OData, SQLite, Power BI, HTML/Excel reporting



Production executive dashboard built from SAP movements by a code-generated Power BI pipeline (figures blurred).

Analytics and planning on S/4HANA



Operations view: daily precast output, cages by type and production by line, group and item (figures blurred).

AI-augmented engineering workflow

Engineering-led development with independent review and automated verification

Building an ERP, design engines and a BIM system next to a full technical-office workload needs leverage without losing control of quality.

What I built

- A multi-agent workflow across Claude Code and OpenAI Codex: work is routed by model, reviewed by a different model family, and released only through automated test gates.
- Safety guards and a single configuration for models, effort and delegation; every decision logged.

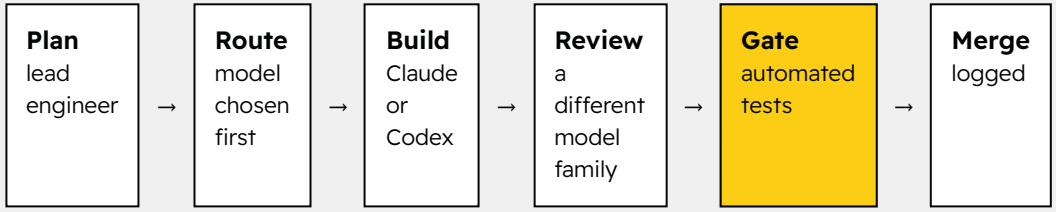
Results

17,000+
automated tests across release gates

2
AI platforms orchestrated

Claude Code, OpenAI Codex, Node.js, Python, Git

How a change reaches production



Controls

- One configuration for models, effort and delegation; every routing decision shown and logged.
- Safety guards block destructive operations; reviewers can never be the producing model family.
- Errors are classified before any retry; unverified runs never count as done.

Engineering software

The tools behind the work in this portfolio, and what each one is used for.

Software	What I do with it	Where it shows
SAP2000 and its API	Finite-element models built and run from code: shells for manholes, a frame on Winkler springs for box culverts, ring frames and 3D solids for RC and jacking pipes, with results read back into the memos.	RC pipes, manholes, box culverts, jacking pipes
Eriksson Pipe	Cross-check of the RC pipe design engine; the comparison agreed within 0.04%.	RC pipes
Eriksson Culvert	Precast box culvert design and checking.	Box culverts
ETABS, SAFE, PROKON	Analysis and design of RC buildings, slabs and foundations, verified by hand to ACI.	Earlier career
Revit 2027 and its C# API	Product systems that generate families, native rebar, views, sheets, BBS and BOM: manholes, RC pipes, jacking pipes and their steel moulds.	BIM, jacking pipes, moulds
Dynamo, AutoCAD	Parametric modelling and DWG issue alongside the Revit sheets.	BIM
SAP S/4HANA Cloud	Material master, BOMs, routings, work centers and classification through Fiori, Migration Cockpit, Mass Maintenance and OData APIs.	SAP, ERP
Python, Excel, VBA	Design engines, memo and catalogue generators, the engineering ERP and the SAP data tools, behind 17,000+ automated tests.	Every engine
Power BI, SQLite	A local SAP data mirror and code-generated production dashboards.	Analytics
Claude Code, OpenAI Codex	Multi-agent development with cross-model review and test gates.	AI workflow

Career, tools and contact

Aug 2023 – Present

Senior Technical Engineer & SAP Master Data Coordinator

Al Jazirah Development Factory for Concrete Products (JDCO), Riyadh

Mar 2023 – Jul 2023

Structural Engineer

United for Telecom Networks Services Company (UTC), Amman

Sep 2022 – Feb 2023

Structural Engineer

Al-Dusuqi Engineering Office, Amman

Jan 2022 – Aug 2022

Structural & Resident Engineer

Al-Majal Engineering Office, Zarqa

2022

B.Sc. Civil Engineering

The Hashemite University, Jordan (ABET-accredited)

Structural analysis: SAP2000 including API automation, Eriksson Culvert, Eriksson Pipe, ETABS, SAFE, PROKON

Standards: ASTM C76, C478, C497, C1433, AASHTO LRFD, ACI 318, MOMRA, BS 5911

SAP: S/4HANA Cloud Public Edition (MM/PP), Fiori, Migration Cockpit, Mass Maintenance, OData APIs, classification and AVC

BIM / CAD: Revit 2027 (families, native rebar, C# API add-ins), Dynamo, AutoCAD, IFC / DWG export

Data & automation: Python, Excel (LET, LAMBDA, XLOOKUP), VBA, SQL/SQLite, Power BI, HTML reporting

AI engineering: Claude Code, OpenAI Codex, multi-agent orchestration, automated verification gates

Languages: Arabic (native), English (professional working proficiency)

Contact
Riyadh, Saudi Arabia
+966 53 726 5921
eyashamadd@gmail.com
Jordanian national